

Tensor Formats in Banach Spaces

Minimal Subspaces, Geometry,
and Variational Principles

Antonio Falcó

ESI International Chair@CEU-UCH

Departamento de Matemáticas, Física y Ciencias Tecnológicas

Universidad Cardenal Herrera-CEU, CEU Universities

San Bartolomé 55, 46115 Alfara del Patriarca (Valencia), Spain

`afalco@uchceu.es`

Ergebnisse der Mathematik und ihrer Grenzgebiete (3. Folge), Springer

May 31, 2026

Contents

Preface	vii
Notation and Conventions	ix
I Algebraic and Topological Foundations	1
1 Introduction and Motivation	3
1.1 The curse of dimensionality	3
1.2 Tensor formats and the role of subspaces	4
1.3 Three fundamental questions	5
1.4 The four source papers	6
1.5 What this monograph adds	6
1.6 Overview of the main results	7
1.7 Guide to the reader	8
1.8 Historical context	9
2 Banach Space Prerequisites	11
2.1 Normed spaces and Banach spaces	11
2.2 Complemented subspaces and projections	13
2.3 The Grassmann–Banach manifold	15
2.4 Banach manifolds	17
2.5 Multilinear maps and Fréchet differentiability	18
3 Algebraic Tensor Spaces	21
3.1 The algebraic tensor product	21
3.2 Matricisation and multilinear rank	23
3.3 The Tucker format	24
3.4 The tensor product map and its differential	26
4 Minimal Subspaces in Tensor Representations	29
4.1 Existence and uniqueness of minimal subspaces	29

4.2	Lattice properties and inclusion under partitions	31
4.3	Admissible ranks and the disjoint decomposition	33
4.4	Best approximations in Banach tensor spaces	34
4.5	Historical notes and comparison with the literature	36
II	Tree-Based Tensor Formats	37
5	Dimension Partition Trees	39
5.1	Definition and basic properties	39
5.2	Special trees and standard formats	41
5.3	Graph-theoretic properties	43
5.4	The intersection representation	44
5.5	Historical notes	45
6	Algebraic Tree-Based Tensor Formats	47
6.1	Tree-based tensor spaces	47
6.2	Tree-based rank and admissible tuples	48
6.3	Minimal subspaces compatible with a tree: new characterisation	49
6.4	The transfer tensor representation	49
6.5	Special cases	51
6.6	Disjoint decomposition and rank stratification	52
6.7	Historical notes and comparison	53
7	Topological Tree-Based Tensor Spaces	55
7.1	Topological tree-based tensor spaces	55
7.2	The T_D -continuity condition	56
7.3	The identification theorem	56
7.4	The tree-based injective norm condition	57
7.5	Weak closedness and best approximations	58
7.6	Historical notes	59
III	Differential Geometry of Tensor Formats	61
8	The Tucker Manifold	63
8.1	The rank map and the decomposition of $\mathbf{V}_D$	63
8.2	Local charts	64
8.3	The $\mathcal{C}^\infty$ -Banach manifold structure	65
8.4	The tangent space	66
8.5	Immersion in the ambient Banach space	67

8.6	The tensor space as a Banach manifold	68
8.7	Historical notes and comparison	68
9	The Tree-Based Manifold	69
9.1	Proper sets of tree-based tensors	69
9.2	The Laplacian-like map and the intersection of model spaces	70
9.3	Charts for the tree-based manifold	71
9.4	The $\mathcal{C}^\infty$ -Banach manifold structure	72
9.5	Tangent space of the tree-based manifold	72
9.6	Immersion in the ambient Banach space	72
9.7	Comparison with alternative descriptions	73
9.8	Topology and connected components	74
9.9	Historical notes	74
IV	Variational Principles and Applications	77
10	The Dirac–Frenkel Variational Principle	79
10.1	The abstract Cauchy problem on tensor Banach spaces	79
10.2	The principle in Hilbert spaces	80
10.3	Projections in Banach spaces	80
10.4	The Dirac–Frenkel principle for the Tucker manifold	81
10.5	Extension to tree-based formats	83
10.6	Historical notes	83
11	Open Problems and Perspectives	85
11.1	Global topology of tensor manifolds	85
11.2	Riemannian and Finsler geometry on tensor manifolds	86
11.3	Algorithms and convergence analysis	86
11.4	Beyond Banach spaces	87
11.5	Connections with quantum information	88
11.6	Learning on tensor manifolds	88
A	Banach Manifolds: A Summary	91
B	Functional Analysis: Selected Results	93
C	Index of Notation	95

Preface

This monograph presents a unified and self-contained treatment of the mathematical foundations of low-rank tensor formats in Banach spaces. It synthesises and extends the results of four research papers by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy, published between 2012 and 2023.

Valencia, May 31, 2026

Antonio Falcó

Notation and Conventions

See the file `notation.tex` for the complete list of macros and symbols used throughout.

Part I

**Algebraic and Topological
Foundations**

Chapter 1

Introduction and Motivation

1.1 The curse of dimensionality

Many of the central problems in modern mathematics and its applications involve functions, operators, or data defined on high-dimensional product domains. A d -variate function $f : I_1 \times \cdots \times I_d \rightarrow \mathbb{R}$ defined on a product of d intervals, for example, lives in a space whose dimension grows exponentially with d : if each factor I_α is discretised with n grid points, the natural approximation space has dimension n^d . For $d = 100$ and $n = 10$ this is 10^{100} , a number that no computer will ever handle directly. This exponential explosion of complexity was famously termed by Bellman the *curse of dimensionality*.

High-dimensional problems of this kind arise throughout science and engineering:

- *Quantum chemistry*: the electronic Schrödinger equation for an N -electron molecule lives in $\mathbb{R}^{3N}$; realistic molecules have N in the hundreds.
- *Uncertainty quantification*: the solution of a PDE with d uncertain parameters is a function on $\mathbb{R}^d \times \Omega$, where Ω is the spatial domain.
- *Stochastic control and finance*: value functions in optimal control problems typically satisfy PDEs in $\mathbb{R}^d$ for large d .
- *Machine learning*: the approximation of unknown functions from data in $\mathbb{R}^d$ is the fundamental task of supervised learning.

The common mathematical structure underlying all these problems is that the object of interest — a function, a quantum state, a probability density — is an element of a *tensor product space* $\bigotimes_{\alpha \in D} V_\alpha$, where $D = \{1, \dots, d\}$ and V_α is a function space on the α -th variable. The curse of dimensionality is the statement that the dimension of this space grows exponentially with $\#(D) = d$.

The strategy for circumventing the curse is to restrict attention to *structured low-rank tensors*: elements of $\mathbf{V}_D$ that can be represented using far fewer parameters than n^d . The systematic study of such representations — their algebraic structure, their topological properties in function spaces, and their geometry as differentiable manifolds — is the subject of this monograph.

1.2 Tensor formats and the role of subspaces

From matrices to tensors

The simplest non-trivial case is $d = 2$: here $\mathbf{V}_D = V_1 \otimes_a V_2$, and a tensor $\mathbf{v} \in \mathbf{V}_D$ corresponds to a linear map (a “matrix”) from V_1^* to V_2 . The *rank* of $\mathbf{v}$ — the minimal r such that $\mathbf{v} = \sum_{\nu=1}^r u_\nu \otimes w_\nu$ — determines the complexity of the representation. In finite dimensions this is the rank of the associated matrix; the Eckart–Young theorem gives the best rank- r approximation in the spectral or Frobenius norm.

The higher-order analogue of rank, however, is not straightforward. For $d \geq 3$ there is no single notion of “tensor rank” with all the good properties of matrix rank. Instead, several notions of rank have proven useful, each corresponding to a different way of decomposing the factors of $\mathbf{V}_D$.

Tucker format

The *Tucker format* [Tuc66] represents a tensor $\mathbf{v}$ as

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C_{(i_\alpha)_{\alpha \in D}}^{(D)} \bigotimes_{\alpha \in D} u_{i_\alpha}^{(\alpha)},$$

where $u_{i_\alpha}^{(\alpha)} \in V_\alpha$ and $C^{(D)} \in \mathbb{R}^{r_1 \times \cdots \times r_d}$ is the *core tensor*. The storage cost is $\sum_\alpha r_\alpha n_\alpha + \prod_\alpha r_\alpha$, which is linear in d when the ranks r_α are bounded. The Tucker format is equivalent to the *multilinear SVD* (HOSVD) of De Lathauwer, De Moor, and Vandewalle [DLDMV00].

The correct functional-analytic description of the Tucker format is through *minimal subspaces*: the minimal subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of $\mathbf{v}$ is the smallest subspace such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a V_{[\alpha]}$ (Chapter 4). The Tucker rank of $\mathbf{v}$ is the tuple $(r_\alpha)_{\alpha \in D}$ where $r_\alpha = \dim U_\alpha^{\min}(\mathbf{v})$.

Hierarchical Tucker and Tensor Train formats

The Tucker format still suffers from the curse of dimensionality through its core tensor: $\prod_\alpha r_\alpha$ grows exponentially with d . The *Hierarchical Tucker* (HT) format [HK09] and

the *Tensor Train* (TT) format [Ose11] resolve this by organising the dimensions into a *hierarchy*, encoded by a rooted tree T_D . Each format restricts the tensor to satisfy rank constraints at every level of the tree, reducing the total storage to $O(dr^3)$ (HT) or $O(dr^2n)$ (TT), where r is the maximal rank.

Tree-based tensor formats

This monograph studies all these formats within a single unified framework: *tree-based Tucker formats*, introduced in [FHN21]. Given a dimension partition tree T_D over D and a rank tuple $\mathbf{r}$, the set

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\text{depth}(T_D)} \mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

of tensors with fixed tree-based rank $\mathbf{r}$ is an intersection of Tucker sets at each level of the tree. The Tucker format is the special case $T_D = T_D^{\text{Tucker}}$ (depth 1); HT and TT correspond to binary trees of depth $\lceil \log_2 d \rceil$ and $d - 1$ respectively.

1.3 Three fundamental questions

The results of this monograph are organised around three fundamental mathematical questions, each corresponding to one part of the book.

Question 1 (Chapter 4). *Does every tensor $\mathbf{v}$ in an infinite-dimensional Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ have a best approximation from the set $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$?*

For convex sets in reflexive Banach spaces, best approximations always exist. The set $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$ is not convex (it is not even a subspace), so the existence of best approximations is non-trivial. The answer, established in Chapters 4 and 7, is *yes*: under a natural condition on the norm (condition (Inj), which holds for all L^p and Sobolev spaces), $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$ is weakly closed in $\mathbf{V}_{D, \|\cdot\|_D}$, and reflexivity gives the best approximation.

Question 2 (Chapters 8 and 9). *Does the set $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ (or $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$) of tensors with fixed rank carry the structure of a differentiable manifold, even in infinite dimensions?*

In finite dimensions, $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is known to be a smooth manifold (a quotient of a matrix group). In infinite dimensions, the standard quotient construction fails because the stabiliser group is infinite-dimensional and the quotient topology may not be well-behaved. The answer, established in [FHN19] for Tucker formats and in [FHN23] for

tree-based formats, is *yes*: an explicit atlas with $\mathcal{C}^\infty$ transition maps can be constructed using the Grassmann–Banach manifold structure of the minimal subspaces.

Question 3 (Chapter 10). *Can the Dirac–Frenkel variational principle — the projection of the vector field onto the tangent space of the manifold — be extended from Hilbert to Banach tensor spaces?*

The answer is *yes*, provided the tangent space $Z^{(D)}(\mathbf{v})$ is a *split subspace* of $\mathbf{V}_{D,\|\cdot\|_D}$ (i.e., has a bounded projection onto it). The immersion theorems of Chapters 8 and 9 — the deepest results of the monograph — establish exactly this.

1.4 The four source papers

This monograph synthesises and expands four research papers written by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy:

[FH2012] *On Minimal Subspaces in Tensor Representations* [FH12]. This paper introduces minimal subspaces as the fundamental algebraic object in tensor representations, proves their existence and uniqueness in general vector spaces, and establishes the existence of best Tucker approximations in reflexive Banach tensor spaces. It constitutes Part I (Chapters 3–4) of this monograph.

[FHN2019] *On the Dirac–Frenkel Variational Principle on Tensor Banach Spaces* [FHN19]. This paper constructs the $\mathcal{C}^\infty$ -Banach manifold structure of $\mathfrak{M}_\tau(\mathbf{V}_D)$ via explicit local charts, proves that the manifold is immersed in $\mathbf{V}_{D,\|\cdot\|_D}$ under condition (Inj), and extends the Dirac–Frenkel variational principle to Banach tensor spaces. It forms the basis of Chapters 8 and 10.

[FHN2021] *Tree-based Tensor Formats* [FHN21]. This paper introduces the general framework of dimension partition trees, proves the existence of best tree-based approximations under conditions weaker than [FH12], and characterises tensors of fixed tree-based rank by means of transfer tensors. It underlies Chapters 5–7.

[FHN2023] *Geometry of Tree-based Tensor Formats in Tensor Banach Spaces* [FHN23]. This paper provides the geometric description of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ as an intersection of Tucker manifolds, proves that it is a $\mathcal{C}^\infty$ -Banach manifold compatible with the Tucker structure at each level of the tree, and extends the Dirac–Frenkel principle to tree-based formats. It forms the basis of Chapter 9.

1.5 What this monograph adds

Beyond collecting and unifying the results of the four source papers, this monograph contributes the following:

1. **Unified notation.** The four papers were written over a decade and use slightly different notational conventions. This monograph adopts a single coherent system (detailed in the Notation chapter), in which the Tucker format is the special case $T_D = T_D^{\text{Tucker}}$ of the tree-based format, minimal subspaces are always denoted $U_\alpha^{\min}(\mathbf{v})$, and the model Banach space of the Tucker manifold is always $\mathbf{E}_D$.
2. **Expanded proofs.** Several results in the papers are stated with proof sketches or references to earlier work. This monograph provides complete proofs of the central theorems, including the full proof of the local chart system (Lemma 8.2.1), the $\mathcal{C}^\infty$ transition maps (Lemma 8.3.1), the tangent space identification (Proposition 8.4.2), and the immersion theorems (Theorems 8.5.3 and 9.6.1).
3. **Self-contained background.** Chapters 2 and 3 provide the Banach space and algebraic tensor product prerequisites in the form and notation used throughout, so that the monograph is accessible without consulting external references for the background.
4. **Historical context and comparison.** Each chapter closes with historical notes that place the results in context and compare them with alternative approaches in the literature.
5. **Open problems.** Chapter 11 collects ten concrete open problems arising naturally from the theory, ranging from the global topology of tensor manifolds to learning theory on tree-based manifolds.

1.6 Overview of the main results

We summarise the principal theorems of the monograph, grouped by theme.

Existence of best approximations (Chapters 4, 7).

- *Theorem 4.4.4* [FH2012]: In a reflexive Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ satisfying (Inj), every $\mathbf{u} \in \mathbf{V}_{D, \|\cdot\|_D}$ has a best Tucker approximation of any prescribed rank.
- *Theorem 7.5.3* [FHN2021]: The same holds for tree-based formats under the weaker local condition (Inj)[T_D].

Manifold structure (Chapters 8, 9).

- *Theorem 8.3.2* [FHN2019]: If the tensor product map is continuous, $\mathfrak{M}_\tau(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$ -Banach manifold modelled on $\mathbf{E}_D = \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times r_\alpha}$. It is a fibre bundle over $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ with fibre $\mathbb{R}_*^{\times r_\alpha}$.

- *Theorem 9.4.1* [FHN2023]: Under (Inj) at each level, $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$ -Banach manifold, with an atlas compatible with the Tucker atlases at each level of T_D .

Immersion theorems (Chapters 8, 9).

- *Theorem 8.5.3* [FHN2019]: Under (Inj), the tangent subspace $Z^{(D)}(\mathbf{v})$ is a split subspace of $\mathbf{V}_{D, \|\cdot\|_D}$, and $\mathfrak{M}_\tau(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D, \|\cdot\|_D}$.
- *Theorem 9.6.1* [FHN2023]: The same holds for $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$, with tangent space $\bigcap_k Z^{(D)}([\mathcal{P}_k(T_D)]\mathbf{v})$.

Dirac–Frenkel variational principle (Chapter 10).

- *Theorem 10.4.1* [FHN2019]: In a reflexive, strictly convex $\mathbf{V}_{D, \|\cdot\|_D}$ satisfying (Inj), the Dirac–Frenkel ODE on $\mathfrak{M}_\tau(\mathbf{V}_D)$ has local solutions, and they are characterised by the metric projection onto $Z^{(D)}(\mathbf{v}_r(t))$.
- *Theorem 10.5.1* [FHN2023]: The same holds on $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$.

1.7 Guide to the reader

The monograph is organised into four parts.

Part I: Algebraic and Topological Foundations (Chapters 2–4). Chapter 2 collects the necessary background from Banach space theory and infinite-dimensional differential geometry. Chapter 3 develops the algebraic theory of tensor products. Chapter 4 is the algebraic core of the monograph: it introduces minimal subspaces, proves their fundamental properties, and establishes the existence of best approximations. *Readers familiar with Banach manifolds may skip Chapter 2 and refer back as needed.*

Part II: Tree-Based Tensor Formats (Chapters 5–7). Chapter 5 introduces dimension partition trees and proves the intersection representation $\mathcal{FT}_\tau(\mathbf{V}_D, T_D) = \bigcap_k \mathfrak{M}_{\tau_k}()$. Chapter 6 develops the algebraic theory of tree-based formats, culminating in the transfer tensor representation (Theorem 6.4.3). Chapter 7 treats the topological theory: the T_D -continuity condition, the identification theorem, and the existence of best tree-based approximations. *Readers primarily interested in the Tucker format may focus on Chapters 4 and then jump to Part III.*

Part III: Differential Geometry of Tensor Formats (Chapters 8–9). These two chapters contain the geometrically deepest results of the monograph. Chapter 8 constructs the $\mathcal{C}^\infty$ Tucker manifold and proves its immersion in $\mathbf{V}_{D, \|\cdot\|_D}$. Chapter 9 extends these results to tree-based formats via the Laplacian map and the intersection of Tucker atlases. *The key technical result is the immersion theorem (Theorems 8.5.3 and 9.6.1), which relies on Lemmas 8.5.1 and 8.5.2.*

Part IV: Variational Principles and Applications (Chapters 10–11). Chapter 10 applies the geometric results to the Dirac–Frenkel variational principle, establishing the existence and characterisation of solutions to the reduced ODE on tensor manifolds. Chapter 11 collects ten open problems.

The three appendices (Appendices A–C) provide quick references for Banach manifold theory, functional analysis, and the notation.

Prerequisites. The reader is assumed to be familiar with functional analysis at the level of a standard graduate course (Banach spaces, weak convergence, Hilbert spaces) and with differential manifolds at the level of a first course using smooth finite-dimensional manifolds. The infinite-dimensional theory (Banach manifolds, Grassmannians) is developed from scratch in Chapter 2.

Notation. A complete list of symbols is given in the Notation chapter following this Introduction. All macros used in the text are defined in the file `notation.tex`.

1.8 Historical context

The mathematical study of tensor approximation has roots in several independent traditions.

In *multilinear algebra*, Hitchcock [Hit27] introduced the notion of tensor rank in 1927. Tucker [Tuc66] introduced the Tucker format in psychometrics in 1966. The HOSVD was introduced by De Lathauwer, De Moor, and Vandewalle [DLDMV00] in 2000.

In *quantum physics*, the matrix product state (MPS) representation appeared in Vidal’s work [Vid03] on quantum simulation and was independently proposed as the TT decomposition by Oseledets and Tyrtshnikov [Ose11]. The DMRG algorithm of White, which is equivalent to optimisation over MPS/TT manifolds, dates from 1992.

In *numerical analysis*, the HT format was introduced by Hackbusch and Kühn [HK09] with the explicit goal of controlling storage costs for high-dimensional PDEs. The existence of best approximations in infinite-dimensional tensor spaces was first established in the present framework by Falcó and Hackbusch [FH12].

The *differential-geometric* treatment of tensor manifolds began with Koch and Lubich [KL07] in the finite-dimensional Hilbert space setting, motivated by the MCTDH method. The extension to infinite-dimensional Banach spaces in [FHN19, FHN23] required new ideas — notably the explicit chart construction via Grassmann charts and the immersion theorem — that form the core of this monograph.

Chapter 2

Banach Space Prerequisites

This chapter collects the functional-analytic and differential-geometric background required throughout the monograph. The reader familiar with Banach space theory and infinite-dimensional differential geometry may proceed directly to Chapter 3, referring back as needed. All vector spaces are over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$; unless stated otherwise, $\mathbb{K} = \mathbb{R}$.

2.1 Normed spaces and Banach spaces

We begin by fixing notation and recalling the central notions of normed and Banach space theory that will be used throughout.

Basic definitions

Let X be a vector space over $\mathbb{K}$. A *norm* on X is a map $\|\cdot\| : X \rightarrow [0, \infty)$ satisfying:

(N1) $\|x\| = 0$ if and only if $x = 0$,

(N2) $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{K}$, $x \in X$,

(N3) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

The pair $(X, \|\cdot\|)$ is called a *normed space*. A sequence $(x_n)_{n \in \mathbb{N}}$ in X is a *Cauchy sequence* if $\|x_n - x_m\| \rightarrow 0$ as $n, m \rightarrow \infty$. A normed space in which every Cauchy sequence converges is called a *Banach space*.

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are *equivalent* if there exist constants $c, C > 0$ such that $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in X$. Equivalent norms define the same topology, and hence the same Cauchy sequences, so they produce the same Banach space structure.

Example 2.1.1. Let $I \subset \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. The Lebesgue space $L^p(I)$, equipped with the norm $\|f\|_{L^p} = \left(\int_I |f(x)|^p dx \right)^{1/p}$, is a Banach space. For

$p = \infty$ one uses the essential supremum norm. The case $p = 2$ gives a Hilbert space with inner product $\langle f, g \rangle = \int_I f(x) \overline{g(x)} dx$.

Example 2.1.2. For a bounded open set $\Omega \subset \mathbb{R}^n$, $1 \leq p < \infty$, and $k \in \mathbb{N}_0$, the Sobolev space $W^{k,p}(\Omega)$ consists of all functions $f \in L^p(\Omega)$ whose distributional derivatives $D^\alpha f$ of order $|\alpha| \leq k$ belong to $L^p(\Omega)$. Equipped with the norm

$$\|f\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p}^p \right)^{1/p},$$

it is a Banach space. For $p = 2$ one writes $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space. These spaces appear as natural component spaces in high-dimensional tensor problems; see Example 4.4.7 and Section 7.5.

Continuous linear maps

Let X and Y be normed spaces. A linear map $A : X \rightarrow Y$ is *continuous* if and only if it is *bounded*, meaning

$$\|A\|_{Y \leftarrow X} := \sup_{x \neq 0} \frac{\|Ax\|_Y}{\|x\|_X} = \sup_{\|x\|_X \leq 1} \|Ax\|_Y < \infty.$$

The quantity $\|A\|_{Y \leftarrow X}$ is the *operator norm* of A . The space of all continuous linear maps from X to Y , denoted $\mathcal{L}(X, Y)$, is itself a normed space under the operator norm; it is a Banach space whenever Y is complete. We write $\mathcal{L}(X) := \mathcal{L}(X, X)$. The set of bounded linear automorphisms (bounded invertible maps with bounded inverse) is

$$\mathrm{GL}(X) := \{A \in \mathcal{L}(X) : A \text{ invertible}\}.$$

This is an open subset of $\mathcal{L}(X)$, and hence a Banach manifold modelled on $\mathcal{L}(X)$; see Example 2.4.3.

Dual spaces and reflexivity

The *topological dual* of a normed space $(X, \|\cdot\|)$ is $X^* := \mathcal{L}(X, \mathbb{K})$, equipped with the dual norm

$$\|\varphi\|_{X^*} := \sup_{\|x\| \leq 1} |\varphi(x)|. \quad (2.1)$$

The dual X^* is always a Banach space, regardless of whether X is complete. The *bidual* $X^{**} := (X^*)^*$ contains X as a canonically embedded subspace via the isometric embedding $\iota : X \rightarrow X^{**}$, $\iota(x)(\varphi) = \varphi(x)$. When ι is surjective, X is called *reflexive*. All L^p spaces with $1 < p < \infty$ are reflexive; L^1 and L^∞ are not.

Reflexivity plays an important role in the existence of best approximations. Recall that a closed set C in a normed space X is *proximal* if every point of X has at least one nearest point in C . In a reflexive Banach space, every closed convex set is proximal (see, e.g., [Hac19, p. 61]). This fact will be used extensively in Chapter 4 to prove the existence of best low-rank tensor approximations.

Convexity and smoothness

Two geometric properties of the norm are of particular importance for the Dirac–Frenkel variational principle in Chapter 10.

Definition 2.1.3. A normed space $(X, \|\cdot\|)$ is *strictly convex* (or *rotund*) if $\|x+y\|/2 < 1$ for all $x, y \in X$ with $\|x\| = \|y\| = 1$ and $x \neq y$.

Definition 2.1.4. A normed space $(X, \|\cdot\|)$ is *uniformly convex* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x+y\|/2 \leq 1 - \delta$ whenever $\|x\|, \|y\| \leq 1$ and $\|x-y\| \geq \varepsilon$.

Every uniformly convex Banach space is reflexive and strictly convex (Clarkson’s theorem). In particular, L^p spaces with $1 < p < \infty$ are uniformly convex. Strict convexity guarantees uniqueness of metric projections onto closed convex sets, a property needed for the well-posedness of the Dirac–Frenkel equation.

Definition 2.1.5. A Banach space X is *smooth* if the limit $\lim_{t \rightarrow 0} (\|x+ty\| - \|x\|)/t$ exists for all $x, y \in X$ with $\|x\| = 1$.

Smoothness is the dual property to strict convexity in the following sense: X is smooth if and only if X^* is strictly convex, and X is strictly convex if and only if X^* is smooth.

2.2 Complemented subspaces and projections

The central geometric notion underlying all manifold structures in this monograph is that of a complemented (or split) subspace. Unlike the Hilbert space setting, complementability is a non-trivial condition in general Banach spaces.

Projections

A bounded linear map $P \in \mathcal{L}(X)$ is a *projection* if $P^2 = P$. In that case $\text{Im } P$ is a closed subspace, $\text{Ker } P$ is closed, and $X = \text{Im } P \oplus \text{Ker } P$ (as a direct sum of closed subspaces). The complementary projection is $\text{id}_X - P$.

Definition 2.2.1. Let X be a Banach space. A closed subspace $U \subset X$ is a *split subspace* (or *complemented subspace*) of X if there exists a closed subspace $W \subset X$ such

that $X = U \oplus W$. The pair (U, W) is then called a *pair of complementary subspaces*, and W is a (*topological*) *complement* of U in X . We write $P_{U \oplus W}$ for the unique continuous projection from X onto U along W .

The following proposition characterises split subspaces in terms of projections and provides three equivalent conditions that will each be used at different points of the text.

Proposition 2.2.2 ([FHHM15, Proposition 4.2]). *Let X be a Banach space and $U \subset X$ a closed subspace. The following conditions are equivalent:*

- (a) U is a split subspace of X .
- (b) There exists $P \in \mathcal{L}(X)$ with $P^2 = P$ and $\text{Im } P = U$.
- (c) There exists $Q \in \mathcal{L}(X)$ with $Q^2 = Q$ and $\text{Ker } Q = U$.

In condition (b), the projection P is none other than $P_{U \oplus W}$ for any choice of closed complement W of U ; in (c), one has $Q = P_{W \oplus U}$.

Proposition 2.2.3 ([FHHM15, Theorem 4.5]). *Let X be a Banach space. Every finite-dimensional subspace of X is a split subspace.*

In a Hilbert space, every closed subspace is complemented (the orthogonal complement provides the split). This is no longer true in general Banach spaces: by a classical result of Lindenstrauss and Tzafriri, a Banach space in which every closed subspace is complemented is necessarily isomorphic to a Hilbert space. The failure of complementability in non-Hilbert spaces is precisely what makes the immersion theorems of Chapter 8 non-trivial.

The nilpotent algebra associated to a splitting

The following result, central to the construction of local charts in Chapters 8 and 9, identifies a particularly well-behaved sub-algebra of $\mathcal{L}(X)$ associated to any splitting $X = U \oplus W$.

Proposition 2.2.4. *Let X be a Banach space and $U, W \in \mathbb{G}(X)$ with $X = U \oplus W$. Define*

$$\mathcal{L}_{(U,W)}(X) := \{P_{W \oplus U} \circ S \circ P_{U \oplus W} : S \in \mathcal{L}(X)\} \cong \mathcal{L}(U, W).$$

Then:

- (i) *The natural map $\mathcal{L}(U, W) \rightarrow \mathcal{L}_{(U,W)}(X)$, $L \mapsto P_{W \oplus U} \circ L \circ P_{U \oplus W}$, is an isometric isomorphism.*

(ii) $\mathcal{L}_{(U,W)}(X)$ is a nilpotent sub-algebra of $\mathcal{L}(X)$: for every $L, L' \in \mathcal{L}_{(U,W)}(X)$ one has $L \circ L' = 0$.

(iii) As a consequence, for every $L \in \mathcal{L}_{(U,W)}(X)$:

$$\exp(L) := \sum_{n=0}^{\infty} \frac{L^n}{n!} = \text{id}_X + L, \quad \exp(-L) = \text{id}_X - L = (\text{id}_X + L)^{-1}. \quad (2.2)$$

Proof. For the isometry in (i), note that $\|P_{W \oplus U} \circ L \circ P_{U \oplus W}\|_{X \leftarrow X} = \|L\|_{W \leftarrow U}$ by the definition of the operator norm. For (ii): if $L = P_{W \oplus U} \circ S \circ P_{U \oplus W}$ and $L' = P_{W \oplus U} \circ S' \circ P_{U \oplus W}$, then

$$L \circ L' = P_{W \oplus U} \circ S \circ (P_{U \oplus W} \circ P_{W \oplus U}) \circ S' \circ P_{U \oplus W} = 0,$$

since $P_{U \oplus W} \circ P_{W \oplus U} = 0$. Statement (iii) follows immediately from (ii) and the series definition of the exponential. $\square$

The identity $\exp(L) = \text{id}_X + L$ means that the exponential map on the nilpotent algebra $\mathcal{L}_{(U,W)}(X)$ is a polynomial of degree one, and in particular a global analytic diffeomorphism. This is the key that makes the local charts of the Grassmann–Banach manifold and the Tucker manifold globally well-defined; see Section 2.3.

Common complements and chart compatibility

The following lemma, which underlies the compatibility of overlapping charts in the Grassmann–Banach manifold, characterises when two split subspaces share a common complement.

Lemma 2.2.5 ([DR63, Lemma 2.1]). *Let X be a Banach space and let $U, U', W \in \mathbb{G}(X)$ with $X = U \oplus W = U' \oplus W$. The following conditions are equivalent:*

- (a) U and U' have W as a common complement, i.e., $X = U \oplus W = U' \oplus W$.
- (b) The restriction $P_{U \oplus W}|_{U'} : U' \rightarrow U$ is a linear isomorphism (with bounded inverse).

Moreover, in this case $(P_{U \oplus W}|_{U'})^{-1} = P_{U' \oplus W}|_U$.

2.3 The Grassmann–Banach manifold

The Grassmannian of a Banach space is the natural parameter space for subspaces, and it serves as the base manifold for the Tucker and tree-based tensor manifolds.

Definition 2.3.1. Let X be a Banach space. The *Grassmann manifold* (or *Grassmannian*) of X is

$$\mathbb{G}(X) := \{U \subset X : U \text{ is a split subspace of } X\}.$$

For $r \in \mathbb{N}_0$, the *finite Grassmannian* is

$$\mathbb{G}_r(X) := \{U \in \mathbb{G}(X) : \dim U = r\}.$$

Both X and $\{0\}$ belong to $\mathbb{G}(X)$. By Proposition 2.2.3, every r -dimensional subspace of X belongs to $\mathbb{G}(X)$, so $\mathbb{G}_r(X)$ is well defined.

Theorem 2.3.2 (Grassmann–Banach manifold, [Dou66]). *Let X be a Banach space. The Grassmannian $\mathbb{G}(X)$ is an analytic Banach manifold, with atlas $\{\mathbb{G}(W, X), \Psi_{U \oplus W}\}_{U \in \mathbb{G}(X)}$, where for each splitting $X = U \oplus W$:*

(i) $\mathbb{G}(W, X) := \{V \in \mathbb{G}(X) : X = V \oplus W\}$ is the chart domain at U .

(ii) The chart map $\Psi_{U \oplus W} : \mathbb{G}(W, X) \rightarrow \mathcal{L}(U, W)$ is defined by

$$\Psi_{U \oplus W}(V) := P_{W \oplus U}|_V \circ (P_{U \oplus W}|_V)^{-1}, \quad (2.3)$$

with inverse $\Psi_{U \oplus W}^{-1}(L) = G(L) := \{(\text{id}_X + L)(u) : u \in U\}$.

The transition maps are analytic. Each chart domain $\mathbb{G}(W, X)$ is a Banach manifold modelled on $\mathcal{L}(U, W)$.

The chart map $\Psi_{U \oplus W}$ sends a subspace $V \in \mathbb{G}(W, X)$ to the linear map $L : U \rightarrow W$ whose graph $G(L)$ equals V . The identity $\Psi_{U \oplus W}^{-1}(L) = \exp(L)(U)$, combined with Proposition 2.2.4, shows that the inverse chart is a degree-one polynomial map (hence analytic). Analyticity of the transition maps follows by a direct computation using Lemma 2.2.5.

Remark 2.3.3. The full Grassmannian $\mathbb{G}(X)$ is a Banach manifold but is *not* modelled on a single Banach space: if U and U' are finite-dimensional with $\dim U \neq \dim U'$, then $\mathcal{L}(U, W)$ and $\mathcal{L}(U', W')$ are non-isomorphic. This is not a deficiency but a structural fact: $\mathbb{G}(X)$ decomposes into connected components $\mathbb{G}(X) = \bigsqcup_{r \in \mathbb{N}_0} \mathbb{G}_r(X)$, each of which is modelled on a single Banach space.

Proposition 2.3.4. *For each $r \in \mathbb{N}_0$, the finite Grassmannian $\mathbb{G}_r(X)$ is a connected component of $\mathbb{G}(X)$, and it is an analytic Banach manifold modelled on $\mathcal{L}(U, W)$ for any $U \in \mathbb{G}_r(X)$ and complement W (the isomorphism class of $\mathcal{L}(U, W)$ does not depend on the choice of U or W).*

Remark 2.3.5 (Grassmannian of a normed space). The construction extends to normed (non-complete) spaces. Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Define $\mathbb{G}_r(X)$

as the set of r -dimensional subspaces of X that are also split in $\bar{X}$. By Lemma 2.5.6, every chart domain $\mathbb{G}(W, \bar{X})$ is contained in $\mathbb{G}_r(X)$, so the atlas of $\mathbb{G}_r(\bar{X})$ restricts to an analytic atlas on $\mathbb{G}_r(X)$. This variant is needed in Section 8.2, where the component spaces V_α are normed but not necessarily complete.

The tangent space to $\mathbb{G}(X)$ at U is identified as follows.

Proposition 2.3.6. *For $U \in \mathbb{G}(X)$ with complement W , the tangent space $T_U \mathbb{G}(X) \cong \mathcal{L}(U, W)$.*

This identification is consistent with the finite-dimensional case: in $\mathbb{R}^n$, the tangent space to the Grassmannian $Gr(r, n)$ at a subspace U is the space of linear maps from U to its orthogonal complement, recovering the classical description.

2.4 Banach manifolds

We recall the definition of a Banach manifold in its general form, since the full Grassmannian $\mathbb{G}(X)$ and the full tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ are manifolds not modelled on a fixed Banach space.

Definition 2.4.1. Let $\mathbb{M}$ be a set. An *atlas of class $\mathcal{C}^p$* (for $p \geq 0$ or $p = \infty$ or $p = \omega$, the analytic case) on $\mathbb{M}$ is a family of *charts* $\{(M_i, u_i)\}_{i \in A}$ satisfying:

(AT1) $\{M_i\}_{i \in A}$ covers $\mathbb{M}$.

(AT2) For each $i \in A$, the chart $u_i : M_i \rightarrow U_i$ is a bijection onto an open set U_i in some Banach space X_i , and for any $i, j \in A$, the set $u_i(M_i \cap M_j)$ is open in X_i .

(AT3) Whenever $M_i \cap M_j \neq \emptyset$, the *transition map* $u_j \circ u_i^{-1} : u_i(M_i \cap M_j) \rightarrow u_j(M_i \cap M_j)$ is a diffeomorphism of class $\mathcal{C}^p$.

Two atlases are *compatible* if their union is again an atlas. A $\mathcal{C}^p$ -*Banach manifold* is a set $\mathbb{M}$ together with a maximal (complete) atlas of class $\mathcal{C}^p$. If all X_i are isomorphic to a single Banach space X , we say $\mathbb{M}$ is *modelled on X* .

Example 2.4.2. Every Banach space X is a $\mathcal{C}^\infty$ -Banach manifold modelled on itself, with the single chart (X, id_X) .

Example 2.4.3. If X is a Banach space, then $\text{GL}(X)$ is an open subset of $\mathcal{L}(X)$, and hence an analytic Banach manifold modelled on $\mathcal{L}(X)$. The maps $A \mapsto A^{-1}$ and $(A, B) \mapsto AB$ are analytic (see [Upm85, 2.7, 2.8]).

Definition 2.4.4. Let $\mathbb{M}$ be a $\mathcal{C}^p$ -Banach manifold with $p \geq 1$ and $m \in \mathbb{M}$. A *tangent vector* at m is an equivalence class of triples (U_i, u_i, v) where (U_i, u_i) is a chart at m and

$v \in X_i$, under the equivalence $(U_i, u_i, v) \sim (U_j, u_j, w)$ if $D(u_j \circ u_i^{-1})(u_i(m))v = w$. The *tangent space* $T_m\mathbb{M}$ is the set of all tangent vectors at m , which inherits the structure of a Banach space from any chart.

Definition 2.4.5. A $\mathcal{C}^1$ -map $f : \mathbb{M} \rightarrow \mathbb{N}$ between Banach manifolds is an *immersion* at m if $Df(m) : T_m\mathbb{M} \rightarrow T_{f(m)}\mathbb{N}$ is injective with closed split image. It is an *immersion* if it is an immersion at every point. An *immersed submanifold* of $\mathbb{N}$ is the image of an injective immersion.

The closedness and split condition on the image of $Df(m)$ is automatic in finite dimensions and in Hilbert spaces (where every closed subspace is split), but is a genuine constraint in Banach spaces. This is the key difficulty resolved in Section 8.5.

2.5 Multilinear maps and Fréchet differentiability

Since the tensor product map is multilinear, the following general results will be used repeatedly.

Definition 2.5.1. A map $M : X_1 \times \cdots \times X_d \rightarrow Y$ between normed spaces is *multilinear* if it is linear in each argument separately. It is *bounded* (equivalently, *continuous*) if

$$\|M\| := \sup_{\substack{x_j \in X_j \\ \|x_j\|_j \leq 1}} \|M(x_1, \dots, x_d)\|_Y < \infty.$$

The following proposition shows that continuity implies a much stronger regularity.

Proposition 2.5.2 ([HN12, Proposition 79]). *Let $X_1, \dots, X_d$ and Y be normed spaces and let $M : X_1 \times \cdots \times X_d \rightarrow Y$ be a continuous multilinear map. Then M is $\mathcal{C}^\infty$ -Fréchet differentiable, and $D^k M = 0$ for all $k > d$.*

The vanishing of derivatives beyond order d reflects the polynomial nature of multilinear maps. In the complex case, a stronger conclusion holds.

Proposition 2.5.3 ([HN12, Theorem 160]). *Let X, Y be complex Banach spaces and $U \subset X$ open. A map $f : U \rightarrow Y$ is analytic if and only if it is $\mathcal{C}^1$ -Fréchet differentiable.*

Corollary 2.5.4. *Every continuous multilinear map between complex Banach spaces is analytic.*

In real Banach spaces, $\mathcal{C}^\infty$ and analytic are distinct notions; the tensor product map is $\mathcal{C}^\infty$ but need not be analytic over $\mathbb{R}$. For the real case, $\mathcal{C}^\infty$ -regularity is sufficient for the manifold constructions in Chapters 8 and 9.

Remark 2.5.5. Proposition 2.5.2 is the reason why the Tucker and tree-based tensor manifolds are $\mathcal{C}^\infty$ -Banach manifolds (and analytic over $\mathbb{C}$): their chart transition maps are compositions of continuous multilinear maps with bounded linear projections, all of which are $\mathcal{C}^\infty$ -Fréchet differentiable.

The Grassmannian revisited: Lemma on normed spaces

For later use in Section 8.2, we record the following lemma about Grassmannians of normed (non-complete) spaces.

Lemma 2.5.6. *Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Let $U \in \mathbb{G}_r(\bar{X})$ with $U \subset X$, and suppose $\bar{X} = U \oplus W$ for some $W \in \mathbb{G}(\bar{X})$. Then every $V \in \mathbb{G}(W, \bar{X})$ satisfies $V \subset X$.*

Proof. Observe that $X = U \oplus (W \cap X)$, where $W \cap X$ is dense in W . Suppose for contradiction that there exists $V \in \mathbb{G}(W, \bar{X})$ with $V \not\subset X$. Since V is finite-dimensional and $\bar{X} = V \oplus W$, the subspace $V \cap X$ is a proper subspace of V . On the other hand, $X = (V \cap X) \oplus (W \cap X)$ implies $\bar{X} = (V \cap X) \oplus W$, so $V \cap X \in \mathbb{G}(W, \bar{X})$. But $\dim(V \cap X) < \dim V = r$, contradicting the fact that every element of $\mathbb{G}(W, \bar{X})$ has dimension r . $\square$

Chapter 3

Algebraic Tensor Spaces

This chapter develops the purely algebraic foundations of tensor spaces. We introduce the algebraic tensor product via its universal property, establish the key notions of elementary tensor, matricisation, and multilinear rank, and define the Tucker format and the set $\mathfrak{M}_t(\mathbf{V}_D)$ of tensors of fixed Tucker rank. No topology is assumed in this chapter; that enters in Chapter 7.

Throughout, $D = \{1, \dots, d\}$ with $d \geq 2$ is the *dimension index set*, and V_α ($\alpha \in D$) are vector spaces over $\mathbb{K}$. Concerning the foundational construction of tensor products we follow Greub [Gre81].

3.1 The algebraic tensor product

Universal property and construction

Definition 3.1.1. The *algebraic tensor product* of the spaces V_α , $\alpha \in D$, is a pair $(\mathbf{V}_D, \otimes)$ consisting of a vector space $\mathbf{V}_D$ and a multilinear map

$$\otimes : \prod_{\alpha \in D} V_\alpha \longrightarrow \mathbf{V}_D, \quad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \quad (3.1)$$

satisfying the following *universal property*: for every vector space Z and every multilinear map $M : \prod_{\alpha \in D} V_\alpha \rightarrow Z$, there exists a unique linear map $\widehat{M} : \mathbf{V}_D \rightarrow Z$ such that $M = \widehat{M} \circ \otimes$.

We write $\mathbf{V}_D = {}_a \otimes_{\alpha \in D} V_\alpha$ and refer to elements of the form $\bigotimes_{\alpha \in D} v_\alpha$ (with $v_\alpha \in V_\alpha$) as *elementary tensors*. Every element of $\mathbf{V}_D$ is a *finite* linear combination of elementary tensors.

The algebraic tensor product exists and is unique up to isomorphism; see [Gre81] for the standard construction. The prefix ‘*a*’ in ${}_a \otimes$ emphasises the algebraic (non-topological) nature. The underlying field is $\mathbb{K} = \mathbb{R}$ throughout, but all results hold equally for $\mathbb{K} = \mathbb{C}$.

Remark 3.1.2. The set of elementary tensors is *not* a basis of $\mathbf{V}_D$; it generates $\mathbf{V}_D$ but with many linear dependencies. For example, $\lambda v \otimes w = v \otimes \lambda w$ for all $\lambda \in \mathbb{K}$. If $\{e_i\}$ and $\{f_j\}$ are bases of V_1 and V_2 respectively, then $\{e_i \otimes f_j\}$ is a basis of $V_1 \otimes_a V_2$.

Bilinear identification and the α -unfolding

For each $\alpha \in D$, the algebraic tensor space can be rewritten as a two-factor product by grouping all indices except α :

$$\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}, \quad \mathbf{V}_\alpha^{[\alpha]} := {}_a \bigotimes_{\beta \in D \setminus \{\alpha\}} V_\beta. \quad (3.2)$$

Concretely, there is a canonical linear isomorphism $\Phi_\alpha : \mathbf{V}_D \xrightarrow{\sim} V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and throughout this monograph we identify $\mathbf{v} \in \mathbf{V}_D$ with $\Phi_\alpha(\mathbf{v}) \in V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ without further comment. An elementary tensor splits as $\bigotimes_{\beta \in D} v_\beta \leftrightarrow v_\alpha \otimes \bigotimes_{\beta \neq \alpha} v_\beta$.

Tensor products of linear maps

Given linear maps $A_\alpha : V_\alpha \rightarrow W_\alpha$ for $\alpha \in D$, their tensor product is the unique linear map

$$\bigotimes_{\alpha \in D} A_\alpha : \mathbf{V}_D \longrightarrow \mathbf{W}_D := {}_a \bigotimes_{\alpha \in D} W_\alpha, \quad (3.3)$$

determined on elementary tensors by $\left(\bigotimes_{\alpha \in D} A_\alpha\right)\left(\bigotimes_{\alpha \in D} v_\alpha\right) = \bigotimes_{\alpha \in D} A_\alpha v_\alpha$. This is the unique linear extension guaranteed by the universal property. We denote the algebraic dual of V by $V' = L(V, \mathbb{K})$ and the space of all linear maps from V to W by $L(V, W)$.

Partitions and iterated tensor products

Definition 3.1.3. Let $\mathcal{P}$ be a partition of D (i.e., a collection of non-empty, pairwise disjoint subsets of D whose union is D). For each $\alpha \in \mathcal{P}$, set $\mathbf{V}_\alpha := {}_a \bigotimes_{j \in \alpha} V_j$ (with the convention $\mathbf{V}_{\{j\}} = V_j$). Then the *iterated tensor product* associated to $\mathcal{P}$ is

$${}_a \bigotimes_{\alpha \in \mathcal{P}} \mathbf{V}_\alpha \cong \mathbf{V}_D, \quad (3.4)$$

where the isomorphism is canonical and consistent with the identifications of elementary tensors.

This associativity-of-tensor-products fact is fundamental: it allows us to regroup the factors of $\mathbf{V}_D$ according to any hierarchical structure, which is precisely what the dimension partition trees of Chapter 5 will exploit.

Example 3.1.4. For $D = \{1, 2, 3, 4\}$ and the partition $\mathcal{P} = \{\{1, 2\}, \{3, 4\}\}$:

$$(V_1 \otimes_a V_2) \otimes_a (V_3 \otimes_a V_4) \cong V_1 \otimes_a V_2 \otimes_a V_3 \otimes_a V_4.$$

The partition $\mathcal{P}' = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ (trivial) recovers the original four-fold product. The non-trivial partition $\mathcal{P}$ is the structure underlying the Tucker format (depth-2 tree), while more refined partitions correspond to hierarchical formats; see Chapter 5.

3.2 Matricisation and multilinear rank

Matricisation in finite dimensions

When the component spaces are finite-dimensional, the tensor $\mathbf{v} \in \mathbf{V}_D$ can be represented as a multi-dimensional array, and the identification (3.2) induces a matrix.

Definition 3.2.1. Let $D = \{1, \dots, d\}$, $\dim V_\alpha = n_\alpha < \infty$, and fix bases of each V_α . Identify $\mathbf{v} \in \mathbf{V}_D$ with its coordinate array $C \in \mathbb{R}^{n_1 \times \dots \times n_d}$. For each $\alpha \in D$, the α -matricisation (or α -unfolding) of $\mathbf{v}$ is the matrix

$$\mathcal{M}_\alpha(\mathbf{v}) \in \mathbb{R}^{n_\alpha \times \prod_{\beta \neq \alpha} n_\beta}$$

obtained by reordering the coordinate array C so that the α -index labels the rows and all other indices, in lexicographic order, label the columns. The rank of this matrix, $\text{rank} \mathcal{M}_\alpha(\mathbf{v}) \in \{0, 1, \dots, n_\alpha\}$, is the α -rank of $\mathbf{v}$.

The notion of α -rank was introduced by Hitchcock [Hit27, p.170] in 1927 under the name “rank on the j th index”. In the infinite-dimensional case, the correct replacement of $\text{rank} \mathcal{M}_\alpha(\mathbf{v})$ is the dimension of the minimal subspace $U_\alpha^{\min}(\mathbf{v})$, which we define in Section 3.3 and develop fully in Chapter 4.

Full-rank core tensors

The following definition isolates the tensors whose coordinate arrays have maximal rank in every direction, a condition that will characterise elements of the Tucker manifold $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$.

Definition 3.2.2. For a rank tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}^d$, define

$$\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha} := \left\{ C \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha} : \text{rank} \mathcal{M}_\alpha(C) = r_\alpha \text{ for all } \alpha \in D \right\}. \quad (3.5)$$

Elements of $\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ are called *full-rank core tensors* of shape $\mathbf{r}$.

Remark 3.2.3. Since $\text{rank } \mathcal{M}_\alpha(C) = r_\alpha$ if and only if $\mathcal{M}_\alpha(C)\mathcal{M}_\alpha(C)^T \in \text{GL}(\mathbb{R}^{r_\alpha})$, and the determinant is a continuous function, $\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ is an open subset of $\mathbb{R}^{\times_{\alpha \in D} r_\alpha}$, hence a smooth finite-dimensional manifold in its own right. In the scalar case $r_\alpha = 1$ for all α , we recover $\mathbb{R}_* = \mathbb{R} \setminus \{0\} = \text{GL}(\mathbb{R})$.

3.3 The Tucker format

Minimal subspaces: first appearance

We introduce minimal subspaces here in their algebraic form; the full theory (existence, uniqueness, lattice properties, best approximation) is developed in Chapter 4.

Definition 3.3.1. Let $\mathbf{v} \in \mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$ and $\alpha \in D$. Under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, the *minimal subspace* of $\mathbf{v}$ in direction α is the unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of smallest dimension such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The α -rank of $\mathbf{v}$ is $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$.

The key properties — existence, uniqueness, and the symmetry $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ — are established in Chapter 4. Here we proceed taking these properties for granted.

The Tucker format and the set $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$

Definition 3.3.2. Let $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ and let $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. A tensor $\mathbf{v} \in \mathbf{V}_D$ is said to be in *Tucker format* with subspace tuple $(U_\alpha)_{\alpha \in D}$ if

$$\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha.$$

The *Tucker rank* of $\mathbf{v}$ is the tuple $\mathbf{r}(\mathbf{v}) = (r_\alpha(\mathbf{v}))_{\alpha \in D}$, and the set of tensors with fixed Tucker rank $\mathbf{r}$ is

$$\mathfrak{M}_\mathbf{r}(\mathbf{V}_D) := \left\{ \mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in D \right\}. \quad (3.6)$$

The Tucker format thus represents $\mathbf{v}$ as an element of a subspace $U_1 \otimes \cdots \otimes U_d$ spanned by r_α -dimensional subspaces $U_\alpha \subset V_\alpha$. Unlike the subspaces, the representation is not unique; the set $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$ collects all tensors for which the *minimal* such subspaces have dimension exactly r_α .

Admissible Tucker ranks

Definition 3.3.3. A tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ is an *admissible Tucker rank* for $\mathbf{V}_D$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $r_\alpha(\mathbf{v}) = r_\alpha$ for all $\alpha \in D$. The set of all admissible Tucker

ranks is

$$\mathcal{AD}(\mathbf{V}_D) := \left\{ (r_\alpha(\mathbf{v}))_{\alpha \in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D \right\}. \quad (3.7)$$

The zero tuple $\mathbf{0} = (0, \dots, 0)$ is always admissible (corresponding to $\mathbf{v} = 0$). The tuple $\mathbf{1} = (1, \dots, 1)$ is admissible if and only if $\dim V_\alpha \geq 1$ for all α . Since the r_α are bounded above by $\dim V_\alpha$ (in finite dimensions), $\mathcal{AD}(\mathbf{V}_D)$ is a finite set when all V_α are finite-dimensional.

The following decomposition is one of the foundational structural facts of the whole theory.

Proposition 3.3.4. *The algebraic tensor space decomposes as a disjoint union:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D). \quad (3.8)$$

Proof. Since every $\mathbf{v} \in \mathbf{V}_D$ has a well-defined Tucker rank tuple (Chapter 4), the sets $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ cover $\mathbf{V}_D$; they are disjoint by definition of the rank. $\square$

The decomposition (3.8) is the algebraic precursor of the manifold decomposition established in Chapter 8: each component $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ will be shown to carry the structure of an infinite-dimensional $\mathcal{C}^\infty$ -Banach manifold.

Explicit representation of Tucker tensors

The following lemma, which combines the definition of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ with the properties of minimal subspaces, gives three equivalent characterisations of Tucker tensors. It is the algebraic heart of the Tucker manifold construction.

Lemma 3.3.5. *Let $\mathbf{V}_D = {}_a \otimes_{\alpha \in D} V_\alpha$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following conditions are equivalent.*

- (a) $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.
- (b) (Tucker representation) *For each $\alpha \in D$ there exists a basis $\{u_{i_\alpha}^{(\alpha)} : 1 \leq i_\alpha \leq r_\alpha\}$ of $U_\alpha^{\min}(\mathbf{v})$ and a unique full-rank core tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$, once the bases are fixed, such that*

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C_{(i_\alpha)_{\alpha \in D}}^{(D)} \bigotimes_{\alpha \in D} u_{i_\alpha}^{(\alpha)}. \quad (3.9)$$

- (c) (Bilinear form) *For each $\alpha \in D$ there exist linearly independent vectors $\{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ in V_α and linearly independent vectors $\{\mathbf{U}_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ in $\mathbf{V}_\alpha^{[\alpha]}$ such that*

$$\mathbf{v} = \sum_{i_\alpha=1}^{r_\alpha} u_{i_\alpha}^{(\alpha)} \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}. \quad (3.10)$$

Moreover, when (3.9) holds, the vectors $\mathbf{U}_{i_\alpha}^{(\alpha)}$ are given by

$$\mathbf{U}_{i_\alpha}^{(\alpha)} = \sum_{\substack{1 \leq i_\beta \leq r_\beta \\ \beta \in D \setminus \{\alpha\}}} C_{i_\alpha, (i_\beta)_{\beta \neq \alpha}}^{(D)} \bigotimes_{\beta \neq \alpha} u_{i_\beta}^{(\beta)}. \quad (3.11)$$

Proof. (a) $\Leftrightarrow$ (c). If (a) holds, then by Proposition 4.1.2 (Chapter 4), $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ with $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) = r_\alpha$, which gives (c). Conversely, (c) implies $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all α , hence (a).

(b) $\Rightarrow$ (c). Immediate.

(c) $\Rightarrow$ (b). From (c), one has $U_\alpha^{\min}(\mathbf{v}) = \text{span}\{u_{i_\alpha}^{(\alpha)}\}$ for each α , and since $\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v})$, there exists $C^{(D)} \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha}$ satisfying (3.9). From (3.11) and the linear independence of the $\mathbf{U}_{i_\alpha}^{(\alpha)}$, one deduces $\text{rank} \mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for all α , so $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$. $\square$

Remark 3.3.6. From Lemma 3.3.5 and Remark 3.2.3, once bases of the minimal subspaces $U_\alpha^{\min}(\mathbf{v})$ are fixed ($\alpha \in D$), there is a natural identification

$$\mathfrak{M}_\tau \left({}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v}) \right) \xrightarrow{\sim} \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}, \quad \mathbf{u} \mapsto E^{(D)}, \quad (3.12)$$

where $E^{(D)}$ is the coordinate array of $\mathbf{u}$ in the chosen bases. This identification is a diffeomorphism (both directions are polynomial), and it is the local model for the Tucker manifold chart; see Section 8.2.

The Tucker format as a flat case of tree-based formats

The Tucker format corresponds to the *shallowest* possible hierarchical structure: the dimension partition tree T_D^{Tucker} of depth 1, whose root is D and whose children are the singletons $\{1\}, \dots, \{d\}$:

$$T_D^{\text{Tucker}} = \{D, \{1\}, \{2\}, \dots, \{d\}\}, \quad S(D) = \{\{1\}, \dots, \{d\}\} = \mathcal{L}(T_D)[T_D^{\text{Tucker}}].$$

In Chapter 6 we will define tree-based formats for arbitrary dimension partition trees T_D and show that $\mathfrak{M}_\tau(\mathbf{V}_D) = \mathcal{FT}_\tau(\mathbf{V}_D, T_D^{\text{Tucker}})$.

3.4 The tensor product map and its differential

When the component spaces are normed, the tensor product map inherits both topological and differential properties that underpin the manifold constructions in Part III.

Definition 3.4.1. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$. The *tensor product map*

$$\bigotimes : \prod_{\alpha \in D} (V_\alpha, \|\cdot\|_\alpha) \longrightarrow (\mathbf{V}_D, \|\cdot\|_D), \quad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \quad (3.13)$$

(where the product is equipped with the max norm $\|(v_\alpha)\|_\times = \max_\alpha \|v_\alpha\|_\alpha$) is said to be *continuous* if $\|\bigotimes_\alpha v_\alpha\|_D \leq C \prod_\alpha \|v_\alpha\|_\alpha$ for some constant $C > 0$.

Continuity of the tensor product map is equivalent to $\|\bigotimes\| < \infty$; it holds whenever $\|\cdot\|_D$ is a crossnorm (Definition 4.4.1) on $\mathbf{V}_D$.

Proposition 3.4.2. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ such that the tensor product map (3.13) is continuous. Then:

(i) The map (3.13) is $\mathcal{C}^\infty$ -Fréchet differentiable, with differential

$$D\bigotimes(v_1, \dots, v_d)(w_1, \dots, w_d) = \sum_{\alpha \in D} v_1 \otimes \dots \otimes v_{\alpha-1} \otimes w_\alpha \otimes v_{\alpha+1} \otimes \dots \otimes v_d. \quad (3.14)$$

(ii) The tensor product of bounded linear maps,

$$\bigotimes : \prod_{\alpha \in D} \mathcal{L}(U_\alpha, V_\alpha) \longrightarrow \mathcal{L}\left(\bigotimes_{\alpha \in D} U_\alpha, \mathbf{V}_D\right), \quad (A_\alpha) \mapsto \bigotimes_{\alpha \in D} A_\alpha,$$

is also continuous and $\mathcal{C}^\infty$ -Fréchet differentiable for any finite-dimensional subspaces $U_\alpha \subset V_\alpha$.

Proof. Part (i) follows directly from Proposition 2.5.2, since $\bigotimes$ is a continuous multilinear map. The formula for the differential follows by the Leibniz rule for multilinear maps. For (ii), one checks boundedness of the map using the continuity of (3.13):

$$\left\| \bigotimes_{\alpha} A_\alpha \right\|_{\mathbf{V}_D \leftarrow \bigotimes U_\alpha} \leq C \prod_{\alpha} \|A_\alpha\|_{V_\alpha \leftarrow U_\alpha},$$

where C is the constant in Definition 3.4.1, and the finite-dimensionality of U_α ensures $L(U_\alpha, V_\alpha) = \mathcal{L}(U_\alpha, V_\alpha)$. Fréchet differentiability then follows again from Proposition 2.5.2. $\square$

Remark 3.4.3. Proposition 3.4.2 is the reason why the Tucker manifold $\mathfrak{M}_t(\mathbf{V}_D)$ is $\mathcal{C}^\infty$ (not merely continuous): all coordinate change maps between charts are built from compositions of projections and the tensor product map, all of which are $\mathcal{C}^\infty$ -Fréchet differentiable. See Section 8.3 for the precise argument.

Banach tensor spaces

When a norm is fixed on $\mathbf{V}_D$, its completion is the natural ambient space.

Definition 3.4.4. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$. The *Banach tensor space* associated to this data is the completion

$$\mathbf{V}_{D, \|\cdot\|_D} := \overline{\|\cdot\|_D} \bigotimes_{\alpha \in D} V_\alpha := \overline{\mathbf{V}_D}^{\|\cdot\|_D}.$$

If $\mathbf{V}_{D, \|\cdot\|_D}$ is a Hilbert space, it is called a *Hilbert tensor space*.

Example 3.4.5. For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 \leq p < \infty$, the Lebesgue spaces $L^p(I_\alpha)$ satisfy

$$L^p\left(\prod_{\alpha \in D} I_\alpha\right) = \|\cdot\|_{0,p} \bigotimes_{\alpha \in D} L^p(I_\alpha),$$

where $\|\cdot\|_{0,p}$ denotes the standard L^p norm on the product domain. For $p = 2$ this is a Hilbert tensor space.

Example 3.4.6. For Sobolev spaces $H^{N,p}(I_\alpha)$ with $1 < p < \infty$ and $\mathbf{I} = \prod_{\alpha} I_\alpha$:

$$H^{N,p}(\mathbf{I}) = \|\cdot\|_{N,p} \bigotimes_{\alpha \in D} H^{N,p}(I_\alpha).$$

This is reflexive and separable. For $p = 2$ it is a Hilbert tensor space. These spaces arise naturally in the numerical treatment of high-dimensional PDEs; see Section 4.4 and Section 7.5.

Remark 3.4.7. The algebraic tensor space $\mathbf{V}_D$ is dense in $\mathbf{V}_{D, \|\cdot\|_D}$ by construction. The Tucker manifold $\mathfrak{M}_t(\mathbf{V}_D)$ lives inside the algebraic space, but its closure (as a subset of $\mathbf{V}_{D, \|\cdot\|_D}$) and its topology as a manifold are determined by the norm $\|\cdot\|_D$ and, in particular, by whether $\|\cdot\|_D$ is a crossnorm. This interplay between algebraic and topological structure is a recurring theme throughout the monograph.

Chapter 4

Minimal Subspaces in Tensor Representations

In Chapter 3 we introduced the Tucker format and defined the minimal subspace $U_\alpha^{\min}(\mathbf{v})$ informally, taking its existence and properties for granted. This chapter provides the complete theory: existence and uniqueness (Section 4.1), the lattice structure and inclusion under refinement of partitions (Section 4.2), admissible ranks and the disjoint decomposition of $\mathbf{V}_D$ (Section 4.3), and the existence of best approximations in Banach tensor spaces (Section 4.4). The results in this chapter are taken from [FH12], expanded with complete proofs and additional examples.

4.1 Existence and uniqueness of minimal subspaces

The case $d = 2$: Schmidt decomposition

The case of two factors $\mathbf{V}_D = V_1 \otimes_a V_2$ is classical and serves as the inductive base. The key observation is a lattice property of tensor subspaces.

Lemma 4.1.1 (Intersection lemma). *Let X_i, Y_i be subspaces of V_i for $i = 1, 2$. Then*

$$(X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2) = (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2). \quad (4.1)$$

Proof. The inclusion $(X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2) \subseteq (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ is immediate. For the reverse inclusion, let $\mathbf{v} \in (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ with representations

$$\mathbf{v} = \sum_{\nu=1}^{n_x} x_\nu^{(1)} \otimes x_\nu^{(2)} = \sum_{\nu=1}^{n_y} y_\nu^{(1)} \otimes y_\nu^{(2)}, \quad x_\nu^{(i)} \in X_i, y_\nu^{(i)} \in Y_i.$$

By adding terms with zero coefficient, assume $n_x = n_y = n$ and that $\{x_\nu^{(1)}\}$ and $\{y_\nu^{(1)}\}$ are each linearly independent (passing to a minimal representation). The two representations yield the same matrix $M_1(\mathbf{v})$ under any identification with a bilinear

form, so $\text{span}\{x_\nu^{(1)}\} = \text{span}\{y_\nu^{(1)}\} \subset X_1 \cap Y_1$. A symmetric argument for the second factor gives $\text{span}\{x_\nu^{(2)}\} \subset X_2 \cap Y_2$, so $\mathbf{v} \in (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2)$. $\square$

Proposition 4.1.2 (Minimal subspace: case $d = 2$). *Let $\mathbf{v} \in V_1 \otimes_a V_2$. There exist unique subspaces $U_i \subset V_i$ of finite dimension such that:*

(i) $\mathbf{v} \in U_1 \otimes_a U_2$.

(ii) If $\mathbf{v} \in X_1 \otimes_a X_2$ for any subspaces X_i , then $U_i \subset X_i$.

Moreover, $\dim U_1 = \dim U_2$.

Proof. Existence. Fix any representation $\mathbf{v} = \sum_\nu v_\nu^{(1)} \otimes v_\nu^{(2)}$ and set $U_i = \text{span}\{v_\nu^{(i)}\}$. By passing to a representation with linearly independent $v_\nu^{(1)}$, one can also take $v_\nu^{(2)}$ linearly independent (otherwise row reduce). Then $\dim U_1 = \dim U_2$ (both equal the rank of the bilinear form).

Minimality. If $\mathbf{v} \in X_1 \otimes_a X_2$, then by the intersection lemma, $\mathbf{v} \in (U_1 \cap X_1) \otimes_a (U_2 \cap X_2)$, which forces $U_i \subset X_i$ by minimality of U_i .

Uniqueness. If U_i and U'_i both satisfy (i)–(ii), minimality of U_i applied to U'_i gives $U_i \subset U'_i$, and vice versa. $\square$

The general case: existence and uniqueness

Theorem 4.1.3 (Existence and uniqueness of minimal subspaces). *Let $D = \{1, \dots, d\}$, $\mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$, and $\mathbf{v} \in \mathbf{V}_D$. For each $\alpha \in D$, under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, there exists a unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of finite dimension $r_\alpha < \infty$ such that:*

(a) $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$.

(b) If $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ for any subspace $U_\alpha \subset V_\alpha$, then $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$.

Furthermore, there exists a unique subspace $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha^{[\alpha]}$ of dimension r_α such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$.

Proof. Apply Proposition 4.1.2 to the two-factor product $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The subspace $U_\alpha^{\min}(\mathbf{v})$ is the minimal subspace in the V_α -factor. Uniqueness follows from the same argument as in $d = 2$. $\square$

Definition 4.1.4. The subspace $U_\alpha^{\min}(\mathbf{v})$ of Theorem 4.1.3 is called the *minimal subspace* of $\mathbf{v}$ in direction α . The integer $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$ is the α -rank of $\mathbf{v}$.

Characterisation via contractions

The following characterisation, which is central to both the proof of Chapter 8 and the tree-based theory of Chapter 6, expresses $U_\alpha^{\min}(\mathbf{v})$ as the range of a family of contractions.

Theorem 4.1.5 (Contraction characterisation). *Let $\mathbf{v} \in \mathbf{V}_D$ and $\alpha \in D$. Then*

$$U_\alpha^{\min}(\mathbf{v}) = \text{span}\left\{(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (\mathbf{V}_\alpha^{[\alpha]})^*\right\}, \quad (4.2)$$

where $(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in V_\alpha$ is the contraction of $\mathbf{v}$ with the functional $\varphi^{[\alpha]}$ in all directions except α .

Proof. Write $\mathbf{v} = \sum_\nu u_\nu \otimes \mathbf{U}_\nu$ with $u_\nu \in V_\alpha$ and $\mathbf{U}_\nu \in \mathbf{V}_\alpha^{[\alpha]}$. For $\varphi \in (\mathbf{V}_\alpha^{[\alpha]})^*$: $(\text{id}_{V_\alpha} \otimes \varphi)(\mathbf{v}) = \sum_\nu \varphi(\mathbf{U}_\nu) u_\nu \in \text{span}\{u_\nu\} = U_\alpha^{\min}(\mathbf{v})$. Conversely, by taking $\varphi = \varphi_\nu$ dual to a basis of $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$, one recovers a basis of $U_\alpha^{\min}(\mathbf{v})$, so equality holds. $\square$

Remark 4.1.6. In Theorem 4.1.5, one can replace $(\mathbf{V}_\alpha^{[\alpha]})^*$ by $(U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}))^*$ without changing the span, since $\mathbf{v} \in V_\alpha \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$. This gives the tighter formula

$$U_\alpha^{\min}(\mathbf{v}) = \text{span}\left\{(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}))^*\right\}, \quad (4.3)$$

which is often more useful in practice.

4.2 Lattice properties and inclusion under partitions

Monotonicity

Proposition 4.2.1 (Monotonicity of minimal subspaces). *Let $\mathbf{v} \in \mathbf{V}_D$ and $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. Then:*

$$\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha \implies U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha \quad \forall \alpha \in D.$$

Proof. Under $\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha$, the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ gives $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, so the minimality of $U_\alpha^{\min}(\mathbf{v})$ (Theorem 4.1.3(b)) yields $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$. $\square$

Inclusion under partition refinement

The following proposition is the key algebraic fact underlying the tree-based format: it shows that minimal subspaces at a coarser level of a tree are contained in tensor products of minimal subspaces at finer levels.

Proposition 4.2.2 (Inclusion under refinement). *Let $\mathbf{v} \in \mathbf{V}_D$ and let $\alpha \subset D$ with $\#(\alpha) \geq 2$. Let $\mathcal{P}_\alpha$ be a non-trivial partition of α . Then*

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v}). \quad (4.4)$$

Proof. From the canonical identification $\mathbf{V}_D \cong \mathbf{V}_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and Theorem 4.1.5,

$$U_\alpha^{\min}(\mathbf{v}) = \text{span} \left\{ (\text{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{\alpha^c}^{\min}(\mathbf{v}))^* \right\}.$$

For each such element $w_\alpha = (\text{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in \mathbf{V}_\alpha$, we apply Proposition 4.2.1 to the tensor space $\mathbf{V}_\alpha \cong {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} \mathbf{V}_\beta$. Since $w_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and $\mathbf{V}_\alpha = {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} \mathbf{V}_\beta$, one has $w_\alpha \in {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(w_\alpha)$. Moreover, from the minimality of $U_\beta^{\min}(\mathbf{v})$ and the relation between contractions at different levels, $U_\beta^{\min}(w_\alpha) \subset U_\beta^{\min}(\mathbf{v})$ for each $\beta \in \mathcal{P}_\alpha$. Hence $w_\alpha \in {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v})$, and taking the span over all such w_α gives (4.4). $\square$

Remark 4.2.3. Proposition 4.2.2 applied iteratively at every internal vertex $\alpha \in T_D$ of a dimension partition tree (Chapter 5) gives the fundamental compatibility of minimal subspaces with the tree structure:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \quad \forall \alpha \in T_D \setminus \mathcal{L}(T_D). \quad (4.5)$$

This is the algebraic foundation of the tree-based format in Chapter 6.

New characterisation of minimal subspaces

The following result, proved in [FHN21] and used in the construction of tree-based tensor formats, gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for $\alpha \supset \beta$.

Proposition 4.2.4. *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \subset D$ with $\#(\alpha) \geq 2$, and $\mathcal{P}_\alpha$ a non-trivial partition of α . Suppose that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in \mathcal{P}_\alpha$ are normed spaces. Then for each $\beta \in \mathcal{P}_\alpha$:*

$$U_\beta^{\min}(\mathbf{v}) = \text{span} \left\{ (\text{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha \setminus \beta)})(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}), \varphi^{(\alpha \setminus \beta)} \in \left({}_a \bigotimes_{\gamma \in \mathcal{P}_\alpha \setminus \{\beta\}} \mathbf{V}_\alpha[\gamma] \right)^* \right\}. \quad (4.6)$$

Proof. Applying Theorem 4.1.5 twice — first for the full index set D and then for α — one obtains two nested contraction characterisations. The formula (4.6) follows by composing these and using that $\mathbf{v} \in \mathbf{V}_\alpha \otimes_a U_{\alpha^c}^{\min}(\mathbf{v})$ allows the replacement of the ambient dual by that of $U_{\alpha^c}^{\min}(\mathbf{v})$. $\square$

4.3 Admissible ranks and the disjoint decomposition

Admissible Tucker ranks

Recall from Chapter 3 (Definition 3.3.3) that the set of admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) = \left\{ (r_\alpha(\mathbf{v}))_{\alpha \in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D \right\}.$$

The following proposition characterises which tuples are admissible.

Proposition 4.3.1. *A tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ belongs to $\mathcal{AD}(\mathbf{V}_D)$ if and only if there exist subspaces $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ and a tensor $\mathbf{v} \in \mathfrak{M}_{\mathbf{r}}(\otimes_{\alpha \in D} U_\alpha)$, i.e., a tensor whose α -unfolding has rank exactly r_α for all $\alpha \in D$.*

Proposition 4.3.2. *For $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

- (i) $r_\alpha \leq \dim V_\alpha$ for all $\alpha \in D$.
- (ii) $r_\alpha = r_{\alpha^c} := \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ for all $\alpha \in D$ (symmetry of marginal ranks).
- (iii) $r_\alpha \leq \prod_{\beta \neq \alpha} r_\beta$ for all $\alpha \in D$.

Proof. Statements (i) and (iii) follow from the representation formula (3.9) and rank properties of multi-index arrays. Statement (ii) follows from Proposition 4.1.2 applied to $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. $\square$

The disjoint decomposition

Theorem 4.1.3 immediately gives the following structural result.

Theorem 4.3.3. *The algebraic tensor space $\mathbf{V}_D$ admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D). \quad (4.7)$$

Each $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is non-empty for $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$. The zero tensor $\mathbf{0}$ belongs to $\mathfrak{M}_{\mathbf{0}}(\mathbf{V}_D)$ and each elementary tensor $\otimes_\alpha v_\alpha$ (with all $v_\alpha \neq 0$) belongs to $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$.

Remark 4.3.4 (Topology of the decomposition). While $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is algebraically well-defined, its topological structure is non-trivial. The key result — that each $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$ -Banach manifold immersed in the ambient Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ — is established in Chapter 8. The immersion requires the condition (Inj) on the norm $\|\cdot\|_D$ (Definition 4.4.1).

Stratification by rank

Definition 4.3.5. For $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, the set of tensors with Tucker rank *bounded by* $\mathfrak{r}$ is

$$\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D) := \bigsqcup_{\substack{\mathfrak{s} \in \mathcal{AD}(\mathbf{V}_D) \\ \mathfrak{s} \leq \mathfrak{r}}} \mathfrak{M}_{\mathfrak{s}}(\mathbf{V}_D) = \left\{ \mathbf{v} \in \mathbf{V}_D : r_{\alpha}(\mathbf{v}) \leq r_{\alpha} \ \forall \alpha \in D \right\}, \quad (4.8)$$

where $\mathfrak{s} \leq \mathfrak{r}$ means $s_{\alpha} \leq r_{\alpha}$ for all $\alpha \in D$.

The set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is not a linear subspace (it is not closed under addition) but it is closed under scalar multiplication. The key property for applications — that best approximations from $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ exist in Banach tensor spaces — is proved in Section 4.4.

4.4 Best approximations in Banach tensor spaces

In numerical applications, one needs to approximate a given tensor $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$ by a tensor of bounded rank. The existence of such a best approximation is non-trivial: the set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is not convex, so general theorems about proximinal sets do not apply. The proof uses the weak closedness of $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$.

Crossnorms and the injective norm condition

Definition 4.4.1. Let $(V_{\alpha}, \|\cdot\|_{\alpha})$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D$. We say $\|\cdot\|_D$ is a *crossnorm* if

$$\left\| \bigotimes_{\alpha \in D} v_{\alpha} \right\|_D = \prod_{\alpha \in D} \|v_{\alpha}\|_{\alpha} \quad \forall v_{\alpha} \in V_{\alpha}. \quad (4.9)$$

The *injective norm* $\|\cdot\|_{\varepsilon}$ on $\mathbf{V}_D$ is defined by

$$\|\mathbf{v}\|_{\varepsilon} := \sup \left\{ \left| \left(\bigotimes_{\alpha \in D} \varphi_{\alpha} \right) (\mathbf{v}) \right| : \varphi_{\alpha} \in V_{\alpha}^*, \ \|\varphi_{\alpha}\|_{V_{\alpha}^*} \leq 1 \right\}.$$

We say $\|\cdot\|_D$ satisfies condition (Inj) if $\|\mathbf{v}\|_D \geq \|\mathbf{v}\|_{\varepsilon}$ for all $\mathbf{v} \in \mathbf{V}_D$.

A crossnorm satisfies (Inj) if and only if it is a *reasonable crossnorm* in the sense that its dual norm is also a crossnorm. The injective norm is the smallest reasonable crossnorm (see [Hac19], Chapter 4).

Remark 4.4.2. The L^p norm on $L^p(\prod_{\alpha} I_{\alpha})$ is a reasonable crossnorm for $1 \leq p \leq \infty$ (see [Hac19], Example 4.72). The Sobolev norm $\|\cdot\|_{(0,1),p}$ on $H^{1,p}(I_1) \otimes_a H^{1,p}(I_2)$ is a crossnorm (Example 4.42 in [Hac19]), hence satisfies (Inj).

Weak closedness of bounded-rank sets

Theorem 4.4.3 (Weak closedness, [FH12, Theorem 5.1]). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying condition (Inj). Then the set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is weakly closed in $\mathbf{V}_{D, \|\cdot\|_D}$.*

Sketch. Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ converge weakly to $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$. We must show $\mathbf{v} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$ for all $\alpha \in D$.

Fix $\alpha \in D$ and consider the α -unfolding map $\mathcal{M}_\alpha : \mathbf{V}_{D, \|\cdot\|_D} \rightarrow \mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$ defined by $\mathcal{M}_\alpha(\mathbf{v})(\varphi) = (\varphi \otimes \text{id})(\mathbf{v})$. Since $r_\alpha(\mathbf{v}_n) \leq r_\alpha$, we have $\text{rank} \mathcal{M}_\alpha(\mathbf{v}_n) \leq r_\alpha$ for all n . The condition (Inj) implies that $\mathcal{M}_\alpha$ is continuous for the weak topology on $\mathbf{V}_{D, \|\cdot\|_D}$ and the weak operator topology on $\mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$. Since the set of operators of rank $\leq r_\alpha$ is closed in the weak operator topology (it is a closed variety), the limit $\mathcal{M}_\alpha(\mathbf{v})$ has rank $\leq r_\alpha$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$. $\square$

Existence of best approximations

Theorem 4.4.4 (Best approximation, [FH12, Theorem 5.3]). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces, let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying (Inj), and assume that $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D, \|\cdot\|_D}$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, there exists $\mathbf{v}_* \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})} \|\mathbf{u} - \mathbf{v}\|_D. \quad (4.10)$$

Proof. Since $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive, any bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian theorem). Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$ be a minimising sequence for the infimum in (4.10). The sequence $(\mathbf{v}_n)$ is bounded (the infimum is attained or approached by a bounded set), so by reflexivity there is a weakly convergent subsequence $\mathbf{v}_{n_k} \rightharpoonup \mathbf{v}_*$. By Theorem 4.4.3, $\mathbf{v}_* \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$. Since the norm is weakly lower semicontinuous, $\|\mathbf{u} - \mathbf{v}_*\|_D \leq \liminf_k \|\mathbf{u} - \mathbf{v}_{n_k}\|_D = \inf$, so $\mathbf{v}_*$ is a best approximation. $\square$

Corollary 4.4.5. *For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 < p < \infty$, the Banach tensor space $L^p(\prod_\alpha I_\alpha)$ is reflexive and its norm satisfies (Inj) (Example 4.4.2). Hence every $f \in L^p(\prod_\alpha I_\alpha)$ has a best Tucker approximation of any prescribed rank.*

Corollary 4.4.6. *For $1 < p < \infty$, $N \geq 1$, and bounded intervals I_α :*

- (i) $H^{N,p}(\prod_\alpha I_\alpha)$ with the standard Sobolev norm is reflexive and satisfies (Inj).
- (ii) The space $H^{1,p}(I_1) \otimes_{\|\cdot\|_{(0,1),p}} H^{1,p}(I_2)$ with the partial-derivative crossnorm satisfies (Inj).

In both cases, best Tucker approximations of any rank exist.

Example 4.4.7. Consider $d = 2$, $V_1 = H^1(0, 1)$, $V_2 = H^1(0, 1)$, and the ambient space $V_1 \otimes_{H^1} V_2$ with the norm $\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|\partial_{x_2} f\|_{L^2}^2$. This Banach tensor space is reflexive and satisfies (Inj), so every $f \in V_1 \otimes_{H^1} V_2$ has a best approximation of rank $\leq r$ for any $r \geq 1$. This setting arises in parametric PDEs where x_1 is the spatial variable and x_2 is the stochastic parameter.

Extension to intersections

In many applications (including the tree-based format), the ambient space is an intersection of Banach tensor spaces, each with its own norm.

Theorem 4.4.8 (Best approximation in intersections, [FH12, Theorem 5.6]). *Let $\mathbf{V}_{D, \|\cdot\|_D}^{(1)}, \dots, \mathbf{V}_{D, \|\cdot\|_D}^{(k)}$ be reflexive Banach tensor spaces over the same algebraic tensor space $\mathbf{V}_D$, each with a norm satisfying (Inj). Equip the intersection $Z = \mathbf{V}_{D, \|\cdot\|_D}^{(1)} \cap \dots \cap \mathbf{V}_{D, \|\cdot\|_D}^{(k)}$ with the norm $\|\mathbf{v}\|_Z = \max_i \|\mathbf{v}\|_i$. If Z is reflexive (which holds, e.g., when $k = 2$ and both norms are reflexive), then every $\mathbf{u} \in Z$ has a best approximation in $\mathfrak{M}_{\leq r}(Z)$.*

4.5 Historical notes and comparison with the literature

The notion of minimal subspace was introduced in [FH12]. It generalises to infinite dimensions the classical concept of matrix rank: for $d = 2$ and finite-dimensional V_α , the α -rank $r_\alpha(\mathbf{v}) = \text{rank} \mathcal{M}_\alpha(\mathbf{v})$ coincides with the classical matrix rank of the α -unfolding, introduced by Hitchcock [Hit27] in 1927. The higher-order version for finite-dimensional spaces was studied by Tucker [Tuc66], De Lathauwer et al. [DLDMV00], and Kolda–Bader [KB09].

The infinite-dimensional theory of [FH12] is motivated by the numerical treatment of high-dimensional PDEs, where $V_\alpha = L^2(\mathbb{R}^3)$ or Sobolev spaces, and the ambient tensor space is infinite-dimensional. The existence of best approximations (Theorem 4.4.4) removes a fundamental obstacle in the convergence analysis of alternating least-squares and related iterative methods.

The weak closedness argument of Theorem 4.4.3 is modelled on Uschmajew’s result for the case of the spectral norm on finite-dimensional spaces [UV13]. The innovation of [FH12] is the extension to general reflexive Banach spaces under the crossnorm condition (Inj), which is the natural condition for tensor product norms.

The connection between minimal subspaces and the geometry of Tucker manifolds is developed in Chapter 8, where the rank map $\varrho : \mathbf{V}_D \rightarrow \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ defined by $\varrho(\mathbf{v}) = (U_\alpha^{\min}(\mathbf{v}))_{\alpha \in D}$ plays a central role in the construction of local charts.

Part II

Tree-Based Tensor Formats

Chapter 5

Dimension Partition Trees

The Tucker format developed in Chapters 3 and 4 corresponds to the shallowest possible hierarchical structure: all d factor spaces are placed at the same level below the root. Richer formats arise by organising the dimensions into a *hierarchy*, encoded by a rooted tree. This chapter introduces dimension partition trees, establishes their basic properties, and presents the principal examples (Tucker, Hierarchical Tucker, Tensor Train). The material prepares the ground for the algebraic (Chapter 6) and topological (Chapter 7) development of tree-based tensor formats.

Throughout, $D = \{1, \dots, d\}$ with $d \geq 2$.

5.1 Definition and basic properties

Definition 5.1.1 (Dimension partition tree). A finite rooted tree T_D is called a *dimension partition tree of D* if:

- (a) Every vertex $\alpha \in T_D$ is a non-empty subset of D .
- (b) The root of T_D is D itself.
- (c) Every non-leaf vertex $\alpha \in T_D$ (i.e., every α with $\#(\alpha) \geq 2$) has at least two children, and the set of children of α , denoted $S(\alpha)$, forms a *non-trivial partition* of α (i.e., a partition into at least two non-empty subsets whose union is α).
- (d) Every vertex α with $\#(\alpha) = 1$ has no children; such vertices are called *leaves*.

The set of all leaves of T_D is denoted $\mathcal{L}(T_D)$. Condition (c) forces every non-leaf to have at least two children, so the tree is never a path of singletons. The following observation is immediate from the definition.

Proposition 5.1.2. *For any dimension partition tree T_D , the set of leaves coincides with the set of singletons of D :*

$$\mathcal{L}(T_D) = \{\{j\} : j \in D\}.$$

In particular, $\mathcal{L}(T_D)$ is the trivial partition of D into singletons.

Proof. By (d), leaves are exactly the vertices with $\#(\alpha) = 1$, i.e., the singletons $\{j\}$ for $j \in D$. Since the root is D and each non-leaf splits into a partition of its label set, every $j \in D$ is eventually reached as a leaf by iterating the partition. $\square$

Levels, depth, and level-partitions

Definition 5.1.3. The *level* of a vertex $\alpha \in T_D$ is defined recursively by

$$\text{level}(D) = 0, \quad \text{level}(\beta) = \text{level}(\alpha) + 1 \quad \text{for } \beta \in S(\alpha).$$

The *depth* of the tree is

$$\text{depth}(T_D) := \max_{\alpha \in T_D} \text{level}(\alpha). \quad (5.1)$$

Since leaves are the only vertices with $\#(\alpha) = 1$, and every path from root to leaf passes through a strictly decreasing chain of subset sizes, the depth satisfies $1 \leq \text{depth}(T_D) \leq d - 1$.

Definition 5.1.4. For $1 \leq k \leq \text{depth}(T_D)$, the *k-th level partition* of D induced by T_D is

$$\mathcal{P}_k(T_D) := \{\alpha \in T_D : \text{level}(\alpha) = k\}. \quad (5.2)$$

By convention, $\mathcal{P}_0(T_D) = \{D\}$.

Proposition 5.1.5. For each $1 \leq k \leq \text{depth}(T_D)$, the collection $\mathcal{P}_k(T_D)$ is a partition of D . Moreover:

(i) Every element of $\mathcal{P}_k(T_D)$ is a subset of some element of $\mathcal{P}_{k-1}(T_D)$ (refinement property).

(ii) $\mathcal{P}_{\text{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$, the trivial partition of D .

Proof. The elements of $\mathcal{P}_k(T_D)$ are precisely the vertex labels at level k . Since each such vertex α is the child of a unique parent $\beta \in \mathcal{P}_{k-1}(T_D)$ satisfying $\alpha \subset \beta$, and since $S(\beta)$ partitions β , the elements of $\mathcal{P}_k(T_D)$ at level k partition D by induction on k . Parts (i) and (ii) follow directly. $\square$

The sequence of partitions $\mathcal{P}_0(T_D) \succ \mathcal{P}_1(T_D) \succ \cdots \succ \mathcal{P}_{\text{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$ (where $\mathcal{P} \succ \mathcal{Q}$ means $\mathcal{Q}$ refines $\mathcal{P}$) captures the hierarchical structure of the tree. This is the central object connecting dimension partition trees to tensor spaces: for each level k , the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ (Equation (3.4)) allows one to view $\mathbf{v} \in \mathbf{V}_D$ as a Tucker tensor at level k .

5.2 Special trees and standard formats

The following three families of dimension partition trees correspond to the three most widely used tensor formats.

The Tucker tree

Example 5.2.1 (Tucker tree). The *Tucker tree* T_D^{Tucker} over D has depth 1 and satisfies

$$T_D^{\text{Tucker}} = \{D, \{1\}, \{2\}, \dots, \{d\}\}, \quad S(D) = \mathcal{L}(T_D)[T_D^{\text{Tucker}}] = \{\{j\} : j \in D\}.$$

The only non-trivial partition step is $D \rightarrow \{\{1\}, \dots, \{d\}\}$.

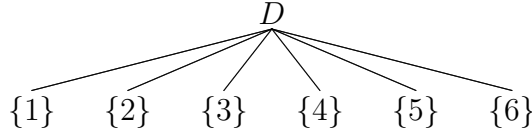


Figure 5.1: Tucker tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 1, one level partition $\mathcal{P}_1 = \mathcal{L}(T_D)$.

The Tucker tree is the unique tree of depth 1 over D . The corresponding tensor format is exactly the Tucker format of Chapter 3: $\mathcal{FT}_r(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_r(\mathbf{V}_D)$.

Binary trees and the Hierarchical Tucker format

Definition 5.2.2. A dimension partition tree T_D is *binary* if every non-leaf vertex has exactly two children: $\#(S(\alpha)) = 2$ for all $\alpha \in T_D$ with $\#(\alpha) \geq 2$.

Binary trees are the natural hierarchical structure for the *Hierarchical Tucker* (HT) format introduced by Hackbusch and Kühn [HK09].

Example 5.2.3 (Hierarchical Tucker tree). For $D = \{1, 2, 3, 4\}$, the balanced binary HT tree is

$$T_D^{\text{HT}} = \{D, \{1, 2\}, \{3, 4\}, \{1\}, \{2\}, \{3\}, \{4\}\},$$

with $S(D) = \{\{1, 2\}, \{3, 4\}\}$, $S(\{1, 2\}) = \{\{1\}, \{2\}\}$, $S(\{3, 4\}) = \{\{3\}, \{4\}\}$. Here $\text{depth}(T_D^{\text{HT}}) = 2$, with level partitions $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D) = \{\{1\}, \{2\}, \{3\}, \{4\}\}$.

Example 5.2.4 (Non-binary tree). For $D = \{1, 2, 3, 4, 5, 6\}$:

$$T_D = \{D, \{1, 2, 3\}, \{4, 5\}, \{6\}, \{1\}, \{2, 3\}, \{4\}, \{5\}, \{2\}, \{3\}\},$$

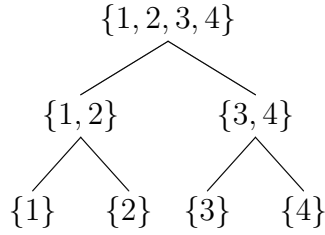


Figure 5.2: Balanced binary tree over $D = \{1, 2, 3, 4\}$: depth 2, corresponding to the HT format.

where $S(D) = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$ (three children), $S(\{1, 2, 3\}) = \{\{1\}, \{2, 3\}\}$, $S(\{2, 3\}) = \{\{2\}, \{3\}\}$, $S(\{4, 5\}) = \{\{4\}, \{5\}\}$. Here $\text{depth}(T_D) = 3$, with $\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$, $\mathcal{P}_2 = \{\{1\}, \{2, 3\}, \{4\}, \{5\}, \{6\}\}$, $\mathcal{P}_3 = \mathcal{L}(T_D)$.

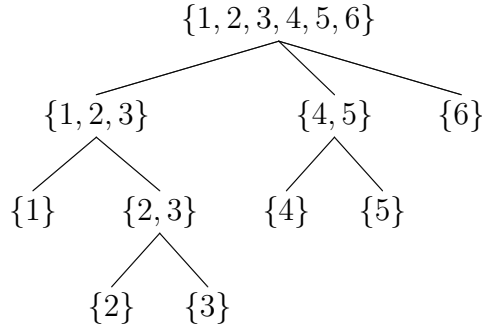


Figure 5.3: Non-binary dimension partition tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 3.

The Tensor Train tree

Example 5.2.5 (Tensor Train tree). For $D = \{1, \dots, d\}$, $d \geq 2$, the *Tensor Train* (TT) tree [Ose11] is the linear (chain) binary tree

$$T_D^{\text{TT}} = \{D\} \cup \{\{j\} : 1 \leq j \leq d\} \cup \{\{1, \dots, j\} : 2 \leq j \leq d-1\},$$

with $S(\{1, \dots, j\}) = \{\{1, \dots, j-1\}, \{j\}\}$ for $j \geq 2$. Here $\text{depth}(T_D^{\text{TT}}) = d-1$. The level partitions are:

$$\mathcal{P}_k(T_D^{\text{TT}}) = \{\{1, \dots, d-k\}, \{d-k+1\}, \dots, \{d\}\}, \quad 1 \leq k \leq d-1.$$

Remark 5.2.6. The TT format is a special case of the HT format: T_D^{TT} is a binary tree with maximal depth $d-1$ and a linear (path) structure, whereas HT trees are typically balanced with depth $\lceil \log_2 d \rceil$. This is reflected in their computational cost: TT has storage $O(dr^2n)$ while balanced HT has $O(dr^3)$, with r the maximal rank and $n = \max_j \dim V_j$ [HK09, Ose11].

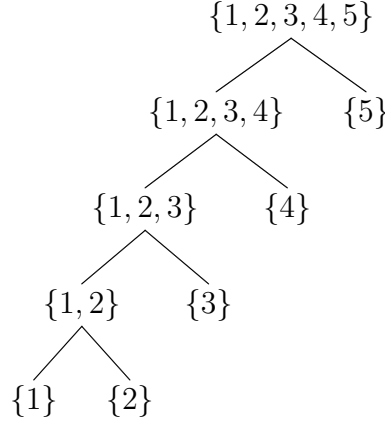


Figure 5.4: Tensor Train tree over $D = \{1, 2, 3, 4, 5\}$: depth 4, linear binary structure.

5.3 Graph-theoretic properties

We record here the combinatorial facts about dimension partition trees that will be used in Chapters 6 and 9.

Proposition 5.3.1. *Let T_D be a dimension partition tree over $D = \{1, \dots, d\}$. Then:*

- (i) $\#(\mathcal{L}(T_D)) = d$. The number of internal (non-leaf) vertices satisfies $\#(T_D \setminus \mathcal{L}(T_D)) \leq d - 1$.
- (ii) If T_D is binary, then $\#(T_D) = 2d - 1$, with exactly $d - 1$ internal vertices.
- (iii) $\text{depth}(T_D) \leq d - 1$, with equality if and only if $T_D = T_D^{\text{TT}}$ (up to re-labelling).

Proof. Part (i): every leaf is a singleton $\{j\}$ for a unique $j \in D$, so $\#(\mathcal{L}(T_D)) = d$. The bound on internal vertices follows since each internal vertex contributes at least two children which are eventually singletons. Part (ii): in a binary tree with d leaves, the number of internal nodes is exactly $d - 1$, giving $2d - 1$ total vertices. Part (iii): the depth is maximised when each step reduces the largest set by exactly one element, giving the TT structure. $\square$

Proposition 5.3.2 (Recursive subtree structure). *Let T_D be a dimension partition tree over D and $\alpha \in T_D$ with $\#(\alpha) \geq 2$. Then the restriction*

$$T_D|_\alpha := \{\beta \in T_D : \beta \subseteq \alpha\}$$

is itself a dimension partition tree over α (with root α).

This recursive structure is fundamental to the inductive proofs in Chapter 6: properties of tree-based tensor formats are proved by induction on $\text{depth}(T_D)$, reducing at each step to the subtrees $T_D|_\beta$ for $\beta \in S(D)$.

5.4 The intersection representation

The central structural theorem of the tree-based theory expresses the set of tree-based tensors with fixed rank as an intersection of Tucker manifolds — one at each level of the tree. This is the key result connecting the tree structure to the geometry of Chapters 9.

For each level partition $\mathcal{P}_k(T_D)$, recall from Section 3.3 that the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ allows one to define the Tucker set

$$\mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) := \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in \mathcal{P}_k(T_D) \right\}, \quad (5.3)$$

where $\mathbf{r}_k = (r_\alpha)_{\alpha \in \mathcal{P}_k(T_D)}$.

Theorem 5.4.1 (Intersection representation, [FHN23, Theorem 3.12]). *Let T_D be a dimension partition tree over D and $\mathbf{r} = (r_\alpha)_{\alpha \in T_D} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then*

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\text{depth}(T_D)} \mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)), \quad (5.4)$$

where $\mathbf{r}_k = (r_\alpha)_{\alpha \in \mathcal{P}_k(T_D)}$.

Sketch. The inclusion $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) \subseteq \bigcap_k \mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ is immediate: if $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all $\alpha \in T_D$, then in particular for all $\alpha \in \mathcal{P}_k(T_D)$ at each level k . For the reverse inclusion, one uses Proposition 4.2.2 inductively: from $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at the leaf level ($k = \text{depth}(T_D)$) and the compatibility condition $U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v})$, one deduces $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at all intermediate levels. See [FHN23] for the complete proof. $\square$

Remark 5.4.2. For $T_D = T_D^{\text{Tucker}}$ (depth 1), the intersection (5.4) reduces to a single factor: $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{L}(T_D)) = \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$, recovering the Tucker format.

Example 5.4.3. For $D = \{1, 2, 3, 4\}$ and the balanced HT tree T_D^{HT} of Example 5.2.3 with $\mathbf{r} = (r_D, r_{\{1,2\}}, r_{\{3,4\}}, r_{\{1\}}, r_{\{2\}}, r_{\{3\}}, r_{\{4\}})$:

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D^{\text{HT}}) = \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{P}_1) \cap \mathfrak{M}_{\mathbf{r}_2}(\mathbf{V}_D, \mathcal{P}_2),$$

where $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D)$. Concretely:

$$\begin{aligned} \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{P}_1) &= \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_{\{1,2\}}^{\min}(\mathbf{v}) = r_{\{1,2\}}, \dim U_{\{3,4\}}^{\min}(\mathbf{v}) = r_{\{3,4\}} \right\}, \\ \mathfrak{M}_{\mathbf{r}_2}(\mathbf{V}_D, \mathcal{P}_2) &= \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_{\{j\}}^{\min}(\mathbf{v}) = r_{\{j\}}, j = 1, 2, 3, 4 \right\}. \end{aligned}$$

The intersection captures tensors that are simultaneously Tucker at the level of pairs ($\{1, 2\}, \{3, 4\}$) and at the level of singletons.

Remark 5.4.4 (Degenerate case). Not all intersections define distinct sets. If for some internal vertex $\alpha \in T_D$ the rank r_α equals $\prod_{\beta \in S(\alpha)} r_\beta$, the rank constraint at α is automatically satisfied whenever the constraints at the children are. In such cases one of the Tucker factors in (5.4) is redundant. This situation is analysed in detail in [FHN23, Theorem 3.12].

5.5 Historical notes

Dimension partition trees formalise the hierarchical structure of tensor representations that had appeared in various forms in the literature. The HT format was introduced by Hackbusch and Kühn in [HK09] using a binary tree structure, with the specific aim of reducing the exponential storage of full Tucker tensors to $O(dr^3)$. The TT format, independently introduced by Vidal [Vid03] in quantum information (under the name *matrix product states*) and by Oseledets [Ose11] in numerical mathematics (as *TT decomposition*), corresponds to the linear chain binary tree.

The general notion of a dimension partition tree, unifying these formats under a common combinatorial framework, was introduced in [FHN21]. The intersection representation (Theorem 5.4.1) was established in [FHN23] and is the key to the geometric theory of Chapter 9: it reduces the analysis of the tree-based manifold $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ to the intersection of Tucker manifolds, each of which has the structure established in Chapter 8.

The connection between dimension partition trees and quantum physics is deeper than a mere formal analogy. In quantum information, a tensor network is a collection of tensors connected by contraction along shared indices, and the tree structure encodes the entanglement structure of a quantum state. The area law for entanglement entropy in one-dimensional quantum systems [Vid03] is the physical motivation for the TT format; balanced binary trees encode area laws in higher-dimensional systems. This connection is discussed further in Chapter 11.

Chapter 6

Algebraic Tree-Based Tensor Formats

This chapter develops the algebraic theory of tree-based tensor formats. Given a dimension partition tree T_D over D and a tensor space $\mathbf{V}_D = {}_a \otimes_{j \in D} V_j$, we associate to each vertex $\alpha \in T_D$ a subspace $\mathbf{V}_\alpha$ and a minimal subspace $U_\alpha^{\min}(\mathbf{v})$, define the tree-based rank and the sets $\mathcal{FT}_r(\mathbf{V}_D, T_D)$ and $\mathcal{FT}_{\leq r}(\mathbf{V}_D, T_D)$, and establish the central characterisation theorem (Theorem 6.4.3) expressing every tensor of fixed tree-based rank in terms of a hierarchical system of *transfer tensors*. The chapter closes with a discussion of the special cases Tucker, HT, and TT.

Throughout, T_D is a dimension partition tree over $D = \{1, \dots, d\}$ with $d \geq 2$, and V_j ($j \in D$) are vector spaces over $\mathbb{K}$.

6.1 Tree-based tensor spaces

Associated spaces at each vertex

For each $\alpha \in T_D$ we associate a vector space $\mathbf{V}_\alpha$ defined recursively from the leaves upward:

- For leaves $\{j\} \in \mathcal{L}(T_D)$: $\mathbf{V}_{\{j\}} = V_j$.
- For internal vertices $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_\alpha = {}_a \otimes_{\beta \in S(\alpha)} \mathbf{V}_\beta$.

Since $S(\alpha)$ partitions α , the canonical identification $\mathbf{V}_\alpha = {}_a \otimes_{j \in \alpha} V_j$ holds at every vertex by the associativity of Equation (3.4). In particular, $\mathbf{V}_D = \mathbf{V}_D$.

Definition 6.1.1. The *tensor space in tree-based format* associated to $(\mathbf{V}_D, T_D)$ is the pair $(\mathbf{V}_D, T_D)$ together with the collection of spaces $\{\mathbf{V}_\alpha\}_{\alpha \in T_D \setminus \{D\}}$.

Minimal subspaces compatible with the tree

Recall from Proposition 4.2.2 that for any $\alpha \in T_D \setminus \mathcal{L}(T_D)$ and any $\mathbf{v} \in \mathbf{V}_D$:

$$U_\alpha^{\min}(\mathbf{v}) \subset_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha. \quad (6.1)$$

This compatibility is the algebraic backbone of the tree-based format: the minimal subspace at each internal vertex is contained in the tensor product of the minimal subspaces of its children.

6.2 Tree-based rank and admissible tuples

Definition 6.2.1 (Tree-based rank). For $\mathbf{v} \in \mathbf{V}_D$, its *tree-based rank with respect to* T_D is the tuple

$$\text{rank}_{T_D}(\mathbf{v}) := \left(\dim U_\alpha^{\min}(\mathbf{v}) \right)_{\alpha \in T_D} \in \mathbb{N}_0^{\#(T_D)}. \quad (6.2)$$

Definition 6.2.2 (Admissible tuples). A tuple $\mathbf{r} = (r_\alpha)_{\alpha \in T_D} \in \mathbb{N}^{\#(T_D)}$ is *admissible for* $(\mathbf{V}_D, T_D)$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $\text{rank}_{T_D}(\mathbf{v}) = \mathbf{r}$. The set of all admissible tuples is

$$\mathcal{AD}(\mathbf{V}_D, T_D) := \left\{ \text{rank}_{T_D}(\mathbf{v}) : \mathbf{v} \in \mathbf{V}_D \right\}. \quad (6.3)$$

Remark 6.2.3. Admissible tuples satisfy the compatibility constraints inherited from Proposition 4.3.2 and Equation (6.1). Specifically, for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$r_\alpha \leq \prod_{\beta \in S(\alpha)} r_\beta. \quad (6.4)$$

Equality in (6.4) is the generic case; strict inequality indicates that the tensor has additional structure at vertex α .

Sets of tensors with fixed and bounded tree-based rank

Definition 6.2.4. Let $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Define:

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) := \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in T_D \right\}, \quad (6.5)$$

$$\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D) := \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) \leq r_\alpha \text{ for all } \alpha \in T_D \right\}. \quad (6.6)$$

The relation between these two sets is:

$$\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D) = \bigsqcup_{\substack{\mathbf{s} \in \mathcal{AD}(\mathbf{V}_D, T_D) \\ \mathbf{s} \leq \mathbf{r}}} \mathcal{FT}_{\mathbf{s}}(\mathbf{V}_D, T_D). \quad (6.7)$$

Remark 6.2.5 (Tucker as special case). For $T_D = T_D^{\text{Tucker}}$ (depth 1, $S(D) = \mathcal{L}(T_D)$), the tree-based rank reduces to the Tucker rank: $\text{rank}_{T_D^{\text{Tucker}}}(\mathbf{v}) = (r_D, r_{\{1\}}, \dots, r_{\{d\}})$. Since $r_D = 1$ for all non-zero $\mathbf{v}$ (the root has a single rank), the effective tuple is $(r_{\{1\}}, \dots, r_{\{d\}})$ and $\mathcal{FT}_\tau(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_\tau(\mathbf{V}_D)$.

6.3 Minimal subspaces compatible with a tree: new characterisation

The following proposition, proved in [FHN21], gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ for an internal vertex β in terms of contractions of $U_\alpha^{\min}(\mathbf{v})$ for its parent $\alpha = \text{parent}(\beta)$. This is the tree-based analogue of the contraction characterisation Theorem 4.1.5.

Proposition 6.3.1 (Tree-based characterisation of minimal subspaces, [FHN21, Proposition 2.3]). *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \in T_D \setminus \mathcal{L}(T_D)$ with $\#(S(\alpha)) \geq 2$, and assume that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in S(\alpha)$ are normed spaces. Then for each $\beta \in S(\alpha)$:*

$$U_\beta^{\min}(\mathbf{v}) = \text{span} \left\{ \left(\text{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha \setminus \beta)} \right) (\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}), \varphi^{(\alpha \setminus \beta)} \in \left(\bigotimes_{\gamma \in S(\alpha) \setminus \{\beta\}} \mathbf{V}_\alpha[\gamma] \right)^* \right\}. \quad (6.8)$$

Proof. This follows directly from Proposition 4.2.4 applied at level k of the tree, with α playing the role of the block and $S(\alpha)$ playing the role of the partition $\mathcal{P}_\alpha$. $\square$

6.4 The transfer tensor representation

The main result of this section is the characterisation of every $\mathbf{v} \in \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ by a system of *transfer tensors* $\{C^{(\alpha)}\}_{\alpha \in T_D}$, one at each vertex. This is the algebraic heart of the tree-based format and the starting point for all computational algorithms (ALS, DMRG, etc.).

Transfer tensors and the representation formula

Definition 6.4.1. Let $\alpha \in T_D$ with $\#(S(\alpha)) \geq 2$ and let $\mathbf{r}_\alpha = (r_\beta)_{\beta \in S(\alpha)}$. A *transfer tensor at vertex α* is an array

$$C^{(\alpha)} \in \mathbb{R}^{r_\alpha \times \prod_{\beta \in S(\alpha)} r_\beta},$$

where we view it as a matrix with r_α rows and $\prod_{\beta \in S(\alpha)} r_\beta$ columns (indexed by the multi-index $(i_\beta)_{\beta \in S(\alpha)}$). The transfer tensor $C^{(\alpha)}$ is said to be *full-rank* if $\text{rank} \mathcal{M}_\alpha(C^{(\alpha)}) = r_\alpha$.

For the root vertex D , the transfer tensor $C^{(D)}$ plays a special role:

Definition 6.4.2. The *root transfer tensor* is

$$C^{(D)} \in \mathbb{R}^{\times_{\alpha \in S(D)} r_\alpha},$$

a multi-dimensional array of shape $\prod_{\alpha \in S(D)} r_\alpha$ (no “row” index since the root has no parent). The full-rank condition at the root is $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in S(D)} r_\alpha}$, the open set of Definition 3.2.2.

The main characterisation theorem

Theorem 6.4.3 (Transfer tensor characterisation, [FHN21, Theorem 2.6]). *Let T_D be a dimension partition tree over D with $\text{depth}(T_D) = L$, and let $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathbf{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following statements are equivalent.*

(a) $\mathbf{v} \in \mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$.

(b) *Given bases $\{u_{i_k}^{(k)} : 1 \leq i_k \leq r_k\}$ of $U_k^{\min}(\mathbf{v})$ for each $k \in \mathcal{L}(T_D)$, there exist:*

- *A unique root transfer tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in S(D)} r_\alpha}$ such that*

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in S(D)}} C_{(i_\alpha)_{\alpha \in S(D)}}^{(D)} \bigotimes_{\alpha \in S(D)} \mathbf{u}_{i_\alpha}^{(\alpha)}, \quad (6.9)$$

- *For each internal vertex $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, a unique transfer tensor $C^{(\mu)} \in \mathbb{R}^{r_\mu \times \prod_{\beta \in S(\mu)} r_\beta}$ with $\text{rank} \mathcal{M}_\mu(C^{(\mu)}) = r_\mu$, and a basis $\{\mathbf{u}_{i_\mu}^{(\mu)} : 1 \leq i_\mu \leq r_\mu\}$ of $U_\mu^{\min}(\mathbf{v})$ given by:*

$$\mathbf{u}_{i_\mu}^{(\mu)} = \sum_{\substack{1 \leq i_\beta \leq r_\beta \\ \beta \in S(\mu)}} C_{i_\mu, (i_\beta)_{\beta \in S(\mu)}}^{(\mu)} \bigotimes_{\beta \in S(\mu)} \mathbf{u}_{i_\beta}^{(\beta)}, \quad 1 \leq i_\mu \leq r_\mu. \quad (6.10)$$

Proof. (b) $\Rightarrow$ (a). Clear: the representation gives $\dim U_\alpha^{\min}(\mathbf{v}) \leq r_\alpha$ at each level, and the full-rank conditions force equality.

(a) $\Rightarrow$ (b). We construct the transfer tensors level by level from the leaves upward.

Step 1 (Root). Since $\mathbf{v} \in \mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v} \in {}_a \bigotimes_{\alpha \in S(D)} U_\alpha^{\min}(\mathbf{v})$ (by the compatibility condition (6.1)), there exists a unique $C^{(D)} \in \mathbb{R}^{\times_{\alpha \in S(D)} r_\alpha}$ such that (6.9) holds (choosing any bases of the $U_\alpha^{\min}(\mathbf{v})$). The full-rank condition $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in S(D)} r_\alpha}$ follows from $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for $\alpha \in S(D)$: the rank $\text{rank} \mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for each α because otherwise one could find a subspace of $U_\alpha^{\min}(\mathbf{v})$ of smaller dimension still containing $\mathbf{v}$, contradicting minimality.

Step 2 (Internal vertices). For each $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, the inclusion $U_\mu^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\mu)} U_\beta^{\min}(\mathbf{v})$ (Proposition 4.2.2) gives the existence of $C^{(\mu)}$

satisfying (6.10). The full-rank condition $\text{rank} \mathcal{M}_\mu(C^{(\mu)}) = r_\mu$ follows from the linear independence of $\{\mathbf{u}_{i_\mu}^{(\mu)}\}$ (as a basis of $U_\mu^{\min}(\mathbf{v})$). $\square$

Remark 6.4.4 (Uniqueness up to gauge). The transfer tensors $\{C^{(\alpha)}\}$ are uniquely determined once the bases $\{u_{i_k}^{(k)}\}_{k \in \mathcal{L}(T_D)}$ are fixed. Changing the bases introduces a *gauge transformation*: if $\{\tilde{u}_{i_k}^{(k)}\}$ is another set of bases related by invertible matrices $G_k \in \text{GL}(\mathbb{R}^{r_k})$ (via $\tilde{u}_{i_k}^{(k)} = \sum_j (G_k)_{ij} u_j^{(k)}$), then the transfer tensors transform by contracting with the corresponding G_k . The freedom to choose orthonormal bases (when the V_j are Hilbert spaces) is exploited in the HOSVD and related algorithms.

Recursive computation

Equation (6.10) has a natural recursive interpretation: the basis vectors of $U_\mu^{\min}(\mathbf{v})$ are built bottom-up from the leaf bases, level by level, using the transfer tensors as “gluing” coefficients. The full representation can be written compactly as:

$$\mathbf{v} = \sum_{(i_j)_{j \in D}} \left[\text{network contraction of } \{C^{(\alpha)}\}_{\alpha \in T_D} \right]_{(i_j)_{j \in D}} \bigotimes_{j \in D} u_{i_j}^{(\{j\})}, \quad (6.11)$$

where the “network contraction” is the tensor network obtained by connecting the transfer tensors along the edges of T_D .

6.5 Special cases

We verify Theorem 6.4.3 in the three standard formats, recovering known representations.

Tucker format

For $T_D = T_D^{\text{Tucker}}$ (depth 1), $S(D) = \mathcal{L}(T_D) = \{\{j\} : j \in D\}$, and there are no internal vertices other than the root. The only transfer tensor is the root tensor $C^{(D)}$, and Theorem 6.4.3 reduces to:

$$\mathbf{v} = \sum_{\substack{1 \leq i_j \leq r_j \\ j \in D}} C_{(i_j)_{j \in D}}^{(D)} \bigotimes_{j \in D} u_{i_j}^{(\{j\})}, \quad C^{(D)} \in \mathbb{R}_*^{\times_{j \in D} r_j}.$$

This is exactly the Tucker representation (3.9) of Lemma 3.3.5, with $C^{(D)}$ playing the role of the core tensor.

Hierarchical Tucker format ($d = 4$, balanced binary)

For T_D^{HT} of Example 5.2.3 with $D = \{1, 2, 3, 4\}$, $S(D) = \{\{1, 2\}, \{3, 4\}\}$, and the two internal vertices $\{1, 2\}$ and $\{3, 4\}$ each have $S(\{i, j\}) = \{\{i\}, \{j\}\}$. Theorem 6.4.3 gives:

- Leaf bases: $\{u_{i_j}^{\{j\}}\}_{i_j=1}^{r_j}$ for $j = 1, 2, 3, 4$.
- Transfer tensors at level 2: $C^{\{1,2\}} \in \mathbb{R}^{r_{\{1,2\}} \times r_1 r_2}$ and $C^{\{3,4\}} \in \mathbb{R}^{r_{\{3,4\}} \times r_3 r_4}$, defining bases of $U_{\{1,2\}}^{\min}(\mathbf{v})$ and $U_{\{3,4\}}^{\min}(\mathbf{v})$:

$$\mathbf{u}_{i_{12}}^{\{1,2\}} = \sum_{i_1=1}^{r_1} \sum_{i_2=1}^{r_2} C_{i_{12}, i_1 i_2}^{\{1,2\}} u_{i_1}^{\{1\}} \otimes u_{i_2}^{\{2\}}.$$

- Root transfer tensor: $C^{(D)} \in \mathbb{R}_*^{r_{\{1,2\}} \times r_{\{3,4\}}}$, giving:

$$\mathbf{v} = \sum_{i_{12}=1}^{r_{\{1,2\}}} \sum_{i_{34}=1}^{r_{\{3,4\}}} C_{i_{12}, i_{34}}^{(D)} \mathbf{u}_{i_{12}}^{\{1,2\}} \otimes \mathbf{u}_{i_{34}}^{\{3,4\}}.$$

The total storage for this representation is $r_1 r_2 r_{\{1,2\}} + r_3 r_4 r_{\{3,4\}} + r_{\{1,2\}} r_{\{3,4\}} + (r_1 + r_2 + r_3 + r_4)n$, where $n = \max_j \dim V_j$, compared to n^4 for a full tensor.

Tensor Train format

For T_D^{TT} over $D = \{1, \dots, d\}$, with $S(\{1, \dots, k\}) = \{\{1, \dots, k-1\}, \{k\}\}$ for $k \geq 2$, Theorem 6.4.3 gives the *TT decomposition*:

$$\mathbf{v} = \sum_{\substack{1 \leq i_j \leq r_j \\ j \in D}} G_{i_1}^{(1)} G_{i_1 i_2}^{(2)} \cdots G_{i_{d-2} i_{d-1}}^{(d-1)} G_{i_{d-1} i_d}^{(d)} \bigotimes_{j \in D} u_{i_j}^{\{j\}}, \quad (6.12)$$

where $G^{(k)} \in \mathbb{R}^{r_{k-1} \times r_k}$ are the transfer matrices at level k (with $r_0 = r_d = 1$). This is the standard TT format of [Ose11], expressed in terms of minimal subspace bases.

6.6 Disjoint decomposition and rank stratification

Theorem 6.6.1. *The algebraic tensor space $\mathbf{V}_D$ admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)} \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D). \quad (6.13)$$

Every non-zero $\mathbf{v} \in \mathbf{V}_D$ belongs to exactly one stratum $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ for $\mathfrak{r} = \text{rank}_{T_D}(\mathbf{v})$.

Proof. Follows directly from the uniqueness of minimal subspaces (Theorem 4.1.3) and the definition of tree-based rank. $\square$

Proposition 6.6.2 (Relation to Tucker decomposition). *For any T_D and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$, the set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is contained in the Tucker set of rank $\mathfrak{r}_{\text{leaf}}$:*

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \subset \mathfrak{M}_{\mathfrak{r}_{\text{leaf}}}(\mathbf{V}_D), \quad \mathfrak{r}_{\text{leaf}} = (r_{\{j\}})_{j \in D}.$$

Moreover, from Theorem 5.4.1:

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) = \mathfrak{M}_{\mathbf{r}_{\text{leaf}}}(\mathbf{V}_D) \cap \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{P}_1(T_D)) \cap \cdots \cap \mathfrak{M}_{\mathbf{r}_{\text{depth}()}-1}(\mathbf{V}_D, \mathcal{P}_{\text{depth}()-1}(T_D)).$$

The tree-based format imposes additional rank constraints beyond the Tucker format at each intermediate level of the tree.

6.7 Historical notes and comparison

The characterisation of tree-based tensors by transfer tensors (Theorem 6.4.3) generalises the Tucker core representation (for depth 1 trees) and the HT representation of Hackbusch–Kühn [HK09] (for binary trees). The TT case (6.12) was established by Oseledets [Ose11]; the connection to the minimal subspace framework was made explicit in [FH12].

The general tree-based format, unifying all these cases, was introduced in [FHN21]. The key innovation there is the *new characterisation of minimal subspaces* (Proposition 6.3.1), which expresses $U_{\beta}^{\min}(\mathbf{v})$ via contractions of $U_{\alpha}^{\min}(\mathbf{v})$ for the parent vertex α . This extends the results of [FH12] (where only the bilinear splitting $V_{\alpha} \otimes V_{\alpha^c}$ was used) to the full tree structure.

The gauge freedom (Remark 6.4.4) is the algebraic manifestation of the non-uniqueness of tensor decompositions. In the TT case, canonical gauges (left-orthogonal, right-orthogonal, and mixed-canonical) are used in the DMRG algorithm [HRS12]. For general trees, the analogous gauge conditions lead to the HOSVD (Higher-Order SVD) of De Lathauwer et al. [DLDMV00].

Chapter 7

Topological Tree-Based Tensor Spaces

The algebraic theory of Chapter 6 operates entirely within the algebraic tensor space $\mathbf{V}_D$. For applications — numerical solution of PDEs, best approximation, dynamical systems — one needs a *topological* framework: a norm on $\mathbf{V}_D$ (or on its completion) and conditions guaranteeing that the tensor product maps at each level of the tree are continuous. This chapter introduces topological tree-based tensor spaces (Section 7.1), the T_D -continuity condition (Section 7.2), the key identification theorem (Section 7.3), and the existence of best approximations in the tree-based setting (Section 7.5). The main results are drawn from [FHN21], which significantly weakens the norm conditions required in [FH12].

7.1 Topological tree-based tensor spaces

Definition 7.1.1 (Topological tree-based tensor space). Let T_D be a dimension partition tree over D and let $(V_j, \|\cdot\|_j)$ be a normed space for each $j \in D$, with completion $V_{j, \|\cdot\|_j}$ (a Banach space). A *topological tree-based representation* of the ambient Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D} = \|\cdot\|_D \otimes_{j \in D} V_j$ is a collection $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}_{\alpha \in T_D \setminus \{D\}}$ such that:

- (i) For each leaf $\{j\} \in \mathcal{L}(T_D)$: $V_{\{j\}, \|\cdot\|_{\{j\}}} = V_{j, \|\cdot\|_j}$.
- (ii) For each internal vertex $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}$ is a Banach tensor space

$$\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}} = \|\cdot\|_{\alpha} \bigotimes_{\beta \in S(\alpha)} V_{\beta}, \quad (7.1)$$

i.e., the completion of the algebraic space $\mathbf{V}_{\alpha} = {}_a \bigotimes_{\beta \in S(\alpha)} \mathbf{V}_{\beta}$ under the norm $\|\cdot\|_{\alpha}$.

The key question is when the norm $\|\cdot\|_{\alpha}$ on the algebraic tensor space ${}_a \bigotimes_{\beta \in S(\alpha)} \mathbf{V}_{\beta}$

is *compatible* with completing each factor: that is, when does

$$\|\cdot\|_\alpha \bigotimes_{\beta \in S(\alpha)} V_{\beta, \|\cdot\|_\beta} = \|\cdot\|_\alpha \bigotimes_{\beta \in S(\alpha)} V_\beta = \|\cdot\|_\alpha \bigotimes_{j \in \alpha} V_j \quad (7.2)$$

hold? The answer is provided by the T_D -continuity condition.

7.2 The T_D -continuity condition

Definition 7.2.1 (T_D -continuous tensor product). Let $\{\mathbf{V}_{\alpha, \|\cdot\|_\alpha}\}_{\alpha \in T_D \setminus \{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D, \|\cdot\|_D}$. The tensor product map $\bigotimes$ is T_D -continuous if for each internal vertex $\alpha \in T_D \setminus \mathcal{L}(T_D)$, the map

$$\bigotimes : \prod_{\beta \in S(\alpha)} (V_{\beta, \|\cdot\|_\beta}, \|\cdot\|_\beta) \longrightarrow (\mathbf{V}_{\alpha, \|\cdot\|_\alpha}, \|\cdot\|_\alpha) \quad (7.3)$$

is continuous (equivalently, bounded as a multilinear map).

T_D -continuity is equivalent to requiring that each norm $\|\cdot\|_\alpha$ is a crossnorm on the product $\prod_{\beta \in S(\alpha)} V_{\beta, \|\cdot\|_\beta}$ (or at least bounds it from above up to a constant).

Lemma 7.2.2 ([Hac19, Lemma 4.34]). *Let $(V_j, \|\cdot\|_j)$ be normed spaces and $\|\cdot\|$ a norm on ${}_a \bigotimes_{j=1}^d V_j$ such that the tensor product map $\bigotimes : \prod_j (V_j, \|\cdot\|_j) \rightarrow ({}_a \bigotimes_j V_j, \|\cdot\|)$ is continuous. Then*

$$\|\cdot\| \bigotimes_{j=1}^d V_{j, \|\cdot\|_j} = \|\cdot\| \bigotimes_{j=1}^d V_j.$$

7.3 The identification theorem

The following theorem is the topological counterpart of the algebraic associativity: it shows that, under T_D -continuity, completing at each vertex of the tree gives the same result as completing all factors simultaneously.

Theorem 7.3.1 (Identification theorem, [FHN21, Theorem 3.2]). *Let $\{\mathbf{V}_{\alpha, \|\cdot\|_\alpha}\}_{\alpha \in T_D \setminus \{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D, \|\cdot\|_D}$ such that the tensor product map $\bigotimes$ is T_D -continuous. Then for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:*

$$\mathbf{V}_{\alpha, \|\cdot\|_\alpha} = \|\cdot\|_\alpha \bigotimes_{\beta \in S(\alpha)} V_{\beta, \|\cdot\|_\beta} = \|\cdot\|_\alpha \bigotimes_{\beta \in S(\alpha)} V_\beta = \|\cdot\|_\alpha \bigotimes_{j \in \alpha} V_j. \quad (7.4)$$

Proof. Apply Lemma 7.2.2 at each internal vertex α , using the T_D -continuity of the tensor product map at α . The third equality in (7.4) follows from the algebraic identity $\mathbf{V}_\alpha = {}_a \bigotimes_{j \in \alpha} V_j$ (associativity of tensor products, Proposition 3.1.3). $\square$

Example 7.3.2 (Anisotropic Sobolev space). Consider $D = \{1, 2, 3\}$, $V_1 = L^p(I_1)$, $V_2 = H^{N,p}(I_2)$, $V_3 = H^{N,p}(I_3)$, and the tree

$$T_D : D \rightarrow \{\{1\}, \{2, 3\}\} \rightarrow \{\{1\}, \{2\}, \{3\}\},$$

with norms $\|\cdot\|_{23}$ on $V_2 \otimes_a V_3$ and $\|\cdot\|_{123}$ on V_D . If both tensor product maps are continuous (i.e., $\otimes$ is T_D -continuous), then

$$\|\cdot\|_{123} \bigotimes_{j=1}^3 V_j = L^p(I_1) \otimes_{\|\cdot\|_{123}} \left(H^{N,p}(I_2) \otimes_{\|\cdot\|_{23}} H^{N,p}(I_3) \right),$$

which is the anisotropic Sobolev space of Figure 7.1.

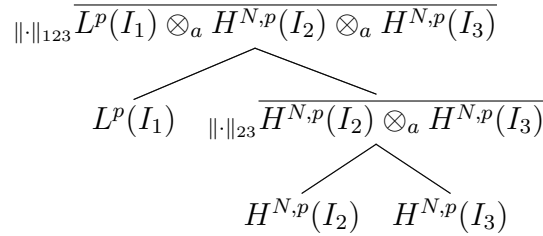


Figure 7.1: Topological tree-based tensor space for an anisotropic Sobolev product. The intermediate space at vertex $\{2, 3\}$ is the Banach completion of $H^{N,p}(I_2) \otimes_a H^{N,p}(I_3)$ under $\|\cdot\|_{23}$.

7.4 The tree-based injective norm condition

For the existence of best approximations, a stronger condition than T_D -continuity is required. The following condition is the tree-based analogue of condition (Inj) from Section 4.4.

Definition 7.4.1 (T_D -injective norm condition). A topological tree-based representation $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}$ satisfies the T_D -injective norm condition (Inj)[T_D] if for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$\|\cdot\|_{\alpha} \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}, \quad (7.5)$$

where $\|\cdot\|_{\varepsilon(S(\alpha))}$ is the injective norm on ${}_a \otimes_{\beta \in S(\alpha)} V_{\beta, \|\cdot\|_{\beta}}$.

Proposition 7.4.2 ([FHN21, Proposition 3.3]). Condition (Inj)[T_D] implies T_D -continuity.

Proof. The injective norm is a reasonable crossnorm (Proposition 4.3.2), so $\|\cdot\|_{\alpha} \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}$ implies that $\otimes : \prod_{\beta \in S(\alpha)} V_{\beta, \|\cdot\|_{\beta}} \rightarrow \mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}$ is continuous. $\square$

Remark 7.4.3. The condition (Inj)[T_D] holds for all common tensor product norms arising from L^p and Sobolev spaces. In particular, the L^p norm on $L^p(\prod_j I_j)$ satisfies

$(\text{Inj})[T_D]$ for any binary tree over D (this follows from the fact that L^p norms are reasonable crossnorms; see [Hac19], Example 4.72).

Minimal subspaces in the completion

Under condition $(\text{Inj})[T_D]$, the definition of minimal subspaces can be extended from $\mathbf{V}_D$ to the Banach completion $\mathbf{V}_{D, \|\cdot\|_D}$. This extension is needed for tensors in $\mathbf{V}_{D, \|\cdot\|_D} \setminus \mathbf{V}_D$, which arise naturally as weak limits of sequences in $\mathbf{V}_D$.

Lemma 7.4.4 (Extension of minimal subspaces, [FH12, Lemma 3.8]). *Let $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}$ satisfy $(\text{Inj})[T_D]$. For $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$ and $\alpha \in S(D)$, define*

$$U_{\alpha}^{\min}(\mathbf{v}) := \overline{\text{span}\left\{(\text{id}_{\mathbf{V}_{\alpha}} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in \bigotimes_{\beta \in S(D) \setminus \{\alpha\}} \mathbf{V}_{\beta}^*\right\}}^{\|\cdot\|_{\alpha}}, \quad (7.6)$$

where $(\text{id}_{\mathbf{V}_{\alpha}} \otimes \varphi^{[\alpha]}) \in \mathcal{L}(\mathbf{V}_{D, \|\cdot\|_D}, \mathbf{V}_{\alpha, \|\cdot\|_{\alpha}})$ is the continuous extension of the contraction map. Then $U_{\alpha}^{\min}(\mathbf{v})$ is a well-defined closed subspace of $\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}$ with $\dim U_{\alpha}^{\min}(\mathbf{v}) < \infty$.

The lemma allows one to define the tree-based rank $\text{rank}_{T_D}(\mathbf{v})$ for any $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$, not just for $\mathbf{v} \in \mathbf{V}_D$. This is essential for the best approximation theorem.

7.5 Weak closedness and best approximations

Theorem 7.5.1 (Weak lower semicontinuity of rank, [FHN21, Theorem 3.4]). *Let $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}$ satisfy $(\text{Inj})[T_D]$, and let $(\mathbf{v}_n) \subset \mathbf{V}_{D, \|\cdot\|_D}$ be a sequence converging weakly to $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$. Then for each $\alpha \in T_D \setminus \{D\}$:*

$$\dim U_{\alpha}^{\min}(\mathbf{v}) \leq \liminf_{n \rightarrow \infty} \dim U_{\alpha}^{\min}(\mathbf{v}_n). \quad (7.7)$$

Sketch. For $\alpha \in S(D)$, the minimal subspace $U_{\alpha}^{\min}(\mathbf{v})$ is characterised as the closure of the range of the operator $\text{id}_{\mathbf{V}_{\alpha}} \otimes \varphi^{[\alpha]}$ applied to $\mathbf{v}$, for $\varphi^{[\alpha]}$ ranging over the dual unit ball. Weak convergence $\mathbf{v}_n \rightharpoonup \mathbf{v}$ implies that $\text{id}_{\mathbf{V}_{\alpha}} \otimes \varphi^{[\alpha]}(\mathbf{v}_n) \rightharpoonup \text{id}_{\mathbf{V}_{\alpha}} \otimes \varphi^{[\alpha]}(\mathbf{v})$ weakly in $\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}$ (by the continuity of the operator from Lemma 7.4.4). The lower semicontinuity of the dimension then follows from the fact that the set of operators of rank $\leq r$ is weakly closed in the operator topology, analogously to Theorem 4.4.3. The extension to all $\alpha \in T_D$ follows by induction on $\text{depth}(T_D)$, using Proposition 6.3.1 to propagate the inequality from children to parents. $\square$

Theorem 7.5.2 (Weak closedness, [FHN21, Theorem 3.5]). *Let $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}$ satisfy $(\text{Inj})[T_D]$ and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ is weakly closed in $\mathbf{V}_{D, \|\cdot\|_D}$.*

Proof. Immediate from Theorem 7.5.1: if $\mathbf{v}_n \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v}_n \rightharpoonup \mathbf{v}$, then $\dim U_{\alpha}^{\min}(\mathbf{v}) \leq \liminf \dim U_{\alpha}^{\min}(\mathbf{v}_n) \leq r_{\alpha}$ for all $\alpha \in T_D$, so $\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$. $\square$

Theorem 7.5.3 (Best approximation, [FHN21, Theorem 3.6]). *Let $\{\mathbf{V}_{\alpha, \|\cdot\|_{\alpha}}\}$ satisfy (Inj)[T_D] and assume $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D, \|\cdot\|_D}$ and $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$, there exists $\mathbf{v}_* \in \mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)} \|\mathbf{u} - \mathbf{v}\|_D. \quad (7.8)$$

Proof. Identical to the proof of Theorem 4.4.4: take a minimising sequence $(\mathbf{v}_n) \subset \mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$, extract a weakly convergent subsequence (by reflexivity), and apply Theorem 7.5.2 to conclude the limit is in $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$. Lower semicontinuity of the norm then gives the minimum. $\square$

Remark 7.5.4 (Improvement over prior results). Theorem 7.5.3 improves the earlier result of Theorem 4.4.4 in two ways:

1. It applies to *any* tree T_D , not just the Tucker tree T_D^{Tucker} .
2. The norm condition (Inj)[T_D] is *weaker* than the condition of [FH12]: it requires the injective norm bound only *locally* at each vertex, whereas [FH12] required a global injective norm bound on the full tensor product.

In particular, Theorem 7.5.3 applies to the anisotropic Sobolev space of Example 7.3.2, which is *not* covered by Theorem 4.4.4.

Corollary 7.5.5. *For $1 < p < \infty$, $L^p(\prod_{j \in D} I_j)$ with any dimension partition tree T_D satisfies the hypotheses of Theorem 7.5.3. Hence every $f \in L^p(\prod_j I_j)$ has a best tree-based approximation of any prescribed tree-based rank.*

7.6 Historical notes

The first best approximation results for tensor formats in infinite-dimensional spaces were proved in [FH12] for the Tucker format, under the assumption that $\|\cdot\|_D \gtrsim \|\cdot\|_{\varepsilon}$ globally. The extension to tree-based formats under the weaker condition (Inj)[T_D] was achieved in [FHN21]. The key improvement is the use of the *local* injective norm bound at each vertex of the tree, combined with the new characterisation of minimal subspaces (Proposition 6.3.1) to propagate weak lower semicontinuity up the tree.

The anisotropic Sobolev space of Example 7.3.2 motivates this improvement: in parametric PDEs where the spatial variable $x_1 \in I_1$ and the parameter $x_2 \in I_2$ have different regularity, the natural norm on $L^p(I_1) \otimes H^{N,p}(I_2)$ is not a global injective norm, but it does satisfy (Inj)[T_D] for the two-level tree. The existence of best approximations in such spaces is crucial for the convergence analysis of greedy algorithms for parametric PDEs (cf. [Nou17]).

Part III

Differential Geometry of Tensor Formats

Chapter 8

The Tucker Manifold

This chapter establishes the central geometric result of the monograph: the set $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ of tensors with fixed Tucker rank carries the structure of a $\mathcal{C}^\infty$ -Banach manifold, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_D, \|\cdot\|_D$. Section 8.1 introduces the rank map and motivates the manifold structure. Section 8.2 constructs the local charts explicitly. Section 8.3 proves the $\mathcal{C}^\infty$ atlas and the fibre bundle structure (Theorem 8.3.2). Section 8.4 identifies the tangent space. Section 8.5 proves the immersion theorem (Theorem 8.5.3). Throughout, $D = \{1, \dots, d\}$, $d \geq 2$, and $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces.

8.1 The rank map and the decomposition of $\mathbf{V}_D$

Recall from Theorem 4.3.3 that

$$\mathbf{V}_D = \bigsqcup_{\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D).$$

The map

$$\varrho : \mathbf{V}_D \longrightarrow \bigcup_{\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)} \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \quad \mathbf{v} \mapsto \left(U_\alpha^{\min}(\mathbf{v}) \right)_{\alpha \in D}, \quad (8.1)$$

sends each tensor to its tuple of minimal subspaces. For fixed $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathbf{r} \neq \mathbf{0}$, the restriction

$$\varrho_{\mathbf{r}} := \varrho|_{\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)} : \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D) \longrightarrow \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha) \quad (8.2)$$

is surjective: for any tuple $(U_\alpha)_{\alpha \in D}$ with $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$, any full-rank tensor in ${}_a \otimes_\alpha U_\alpha$ belongs to $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ and maps to (U_α) .

The product of Grassmannians $\prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha)$ is a Banach manifold (Theorem 2.3.2 applied factor by factor), and the rank map $\varrho_{\mathbf{r}}$ is its natural base for the bundle structure of $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$.

8.2 Local charts

Fix $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathbf{r} \neq \mathbf{0}$ and $\mathbf{v} \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$. Let $V_{\alpha, \|\cdot\|_{\alpha}}$ denote the completion of V_{α} and choose for each $\alpha \in D$ a closed complement $W_{\alpha}^{\min}(\mathbf{v})$ of $U_{\alpha}^{\min}(\mathbf{v})$ in $V_{\alpha, \|\cdot\|_{\alpha}}$:

$$V_{\alpha, \|\cdot\|_{\alpha}} = U_{\alpha}^{\min}(\mathbf{v}) \oplus W_{\alpha}^{\min}(\mathbf{v}).$$

Such a complement exists by Proposition 2.2.3, since $U_{\alpha}^{\min}(\mathbf{v})$ is finite-dimensional.

Chart domain

Define the *chart domain at $\mathbf{v}$* by

$$\mathcal{U}(\mathbf{v}) := \left\{ \mathbf{w} \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D) : U_{\alpha}^{\min}(\mathbf{w}) \in \mathbb{G}(W_{\alpha}^{\min}(\mathbf{v}), V_{\alpha}), \forall \alpha \in D \right\}, \quad (8.3)$$

where $\mathbb{G}(W, X) = \{U \in \mathbb{G}(X) : X = U \oplus W\}$ is the chart domain of the Grassmannian at $U_{\alpha}^{\min}(\mathbf{v})$ (see Section 2.3).

Chart map

Fix bases $\{u_{i_{\alpha}}^{(\alpha)}\}_{i_{\alpha}=1}^{r_{\alpha}}$ of $U_{\alpha}^{\min}(\mathbf{v})$ for each $\alpha \in D$. By Lemma 3.3.5, there is a unique $C^{(D)}(\mathbf{v}) \in \mathbb{R}_*^{\times r_{\alpha}}$ such that $\mathbf{v}$ has the Tucker representation (3.9). For each $\mathbf{w} \in \mathcal{U}(\mathbf{v})$, define the *chart map*

$$\xi_{\mathbf{v}} : \mathcal{U}(\mathbf{v}) \longrightarrow \prod_{\alpha \in D} \mathcal{L}(U_{\alpha}^{\min}(\mathbf{v}), W_{\alpha}^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_{\alpha}}, \quad (8.4)$$

by sending $\mathbf{w} = (\otimes_{\alpha} \exp(L_{\alpha}))(\mathbf{u}(E^{(D)}))$ to the pair $((L_{\alpha})_{\alpha \in D}, E^{(D)})$, where $(L_{\alpha}, E^{(D)})$ are uniquely determined by Lemma 8.2.1 below.

Lemma 8.2.1 (Local representation, [FHN19, Lemma 3.7]). *Under the conditions above, for $\mathbf{w} \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ the following are equivalent:*

- (a) $\mathbf{w} \in \mathcal{U}(\mathbf{v})$.
- (b) *There exists a unique pair $((L_{\alpha})_{\alpha \in D}, E^{(D)}) \in \prod_{\alpha} \mathcal{L}(U_{\alpha}^{\min}(\mathbf{v}), W_{\alpha}^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_{\alpha}}$ such that*

$$\mathbf{w} = \left(\bigotimes_{\alpha \in D} \exp(L_{\alpha}) \right) (\mathbf{u}(E^{(D)})), \quad (8.5)$$

where $\mathbf{u}(E^{(D)}) \in \mathfrak{M}_{\mathbf{r}}((\cdot)_a \otimes_{\alpha} U_{\alpha}^{\min}(\mathbf{v}))$ is the tensor with core $E^{(D)}$ in the fixed bases.

Here $\exp(L_{\alpha}) = \text{id}_{V_{\alpha}} + L_{\alpha}$ by the nilpotency of $L_{\alpha} \in \mathcal{L}(U_{\alpha}^{\min}(\mathbf{v}), W_{\alpha}^{\min}(\mathbf{v}))(V_{\alpha})$ (Proposition 2.2.4), so the representation is polynomial and $\mathcal{C}^{\infty}$.

Remark 8.2.2. Formula (8.5) has a clean geometric interpretation: $\mathbf{w}$ is obtained from the “reference” tensor $\mathbf{u}(E^{(D)})$ (which lives in the fixed subspaces $U_\alpha^{\min}(\mathbf{v})$) by applying the “rotation” $\bigotimes_\alpha \exp(L_\alpha)$ which moves each subspace $U_\alpha^{\min}(\mathbf{v})$ to $U_\alpha^{\min}(\mathbf{w}) = G(L_\alpha) = \text{span}\{(\text{id} + L_\alpha)(u_{i_\alpha}^{(\alpha)})\}$.

8.3 The $\mathcal{C}^\infty$ -Banach manifold structure

Lemma 8.3.1 (Smooth transitions, [FHN19, Lemma 3.8]). *Under the continuity assumption on the tensor product map (3.13), for $\mathbf{v}, \mathbf{v}' \in \mathfrak{M}_\tau(\mathbf{V}_D)$ with $\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}') \neq \emptyset$, the transition map*

$$\xi_{\mathbf{v}'} \circ \xi_{\mathbf{v}}^{-1} : \xi_{\mathbf{v}}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')) \longrightarrow \xi_{\mathbf{v}'}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}'))$$

is $\mathcal{C}^\infty$ -Fréchet differentiable (analytic if the V_α are complex Banach spaces).

Sketch. For $\mathbf{w} \in \mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')$, the transition map is

$$(L_\alpha, E^{(D)}) \mapsto \left((\Psi_{\mathbf{v}'}^{(\alpha)} \circ (\Psi_{\mathbf{v}}^{(\alpha)})^{-1})(L_\alpha), f(\mathfrak{L}, \mathbf{u}(E^{(D)})) \right),$$

where $\Psi_{\mathbf{v}}^{(\alpha)}$ is the Grassmann chart map at $U_\alpha^{\min}(\mathbf{v})$ (Theorem 2.3.2), and $f(\mathfrak{L}, \mathbf{u}(E^{(D)})) = (\bigotimes_\alpha \exp(L_\alpha - L'_\alpha))(\mathbf{u}(E^{(D)}))$. The first component is analytic (transition map of the Grassmannian). The second is $\mathcal{C}^\infty$ by Proposition 3.4.2 (tensor product of $\mathcal{C}^\infty$ maps) and the fact that the evaluation map is multilinear and continuous. See [FHN19] for the complete argument. $\square$

Theorem 8.3.2 ($\mathcal{C}^\infty$ Tucker manifold, [FHN19, Theorem 3.17]). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces for $\alpha \in D$ and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ such that the tensor product map (3.13) is continuous. Then:*

- (i) *The collection $\{(\mathcal{U}(\mathbf{v}), \xi_{\mathbf{v}})\}_{\mathbf{v} \in \mathfrak{M}_\tau(\mathbf{V}_D)}$ is a $\mathcal{C}^\infty$ -atlas on $\mathfrak{M}_\tau(\mathbf{V}_D)$, which is thus a $\mathcal{C}^\infty$ -Banach manifold modelled on the Banach space*

$$\mathbf{E}_D := \prod_{\alpha \in D} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times_{\alpha \in D} r_\alpha}. \quad (8.6)$$

- (ii) *The triple $(\mathfrak{M}_\tau(\mathbf{V}_D), \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \varrho_\tau)$ is a $\mathcal{C}^\infty$ -fibre bundle with typical fibre $\mathbb{R}_*^{\times r_\alpha}$.*

If the V_α are complex Banach spaces, the manifold is analytic.

Proof. The atlas conditions AT1–AT3 are verified as follows. AT1: $\{\mathcal{U}(\mathbf{v})\}$ covers $\mathfrak{M}_\tau(\mathbf{V}_D)$ since every $\mathbf{v}$ belongs to its own chart domain. AT2: $\xi_{\mathbf{v}}$ is a bijection onto $\prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_\alpha}$ by Lemma 8.2.1. AT3: Lemma 8.3.1 gives $\mathcal{C}^\infty$

transitions. The fibre bundle structure follows because $\varrho_{\mathbf{t}}$ acts locally as the projection onto $\prod_{\alpha} \mathbb{G}(W_{\alpha}^{\min}(\mathbf{v}), V_{\alpha})$ (the Grassmannian chart domain), with fibre $\mathbb{R}_*^{\times r_{\alpha}}$. $\square$

Remark 8.3.3 (Independence from the norm). The manifold structure of $\mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D)$ is independent of the choice of $\|\cdot\|_D$: the chart maps $\xi_{\mathbf{v}}$ depend only on the splitting $V_{\alpha, \|\cdot\|_{\alpha}} = U_{\alpha}^{\min}(\mathbf{v}) \oplus W_{\alpha}^{\min}(\mathbf{v})$, and the $\mathcal{C}^{\infty}$ regularity follows from the continuity of the tensor product map, which holds for any crossnorm. Different norms give equivalent smooth structures.

Corollary 8.3.4 (Hilbert manifold). *If $V_{\alpha, \|\cdot\|_{\alpha}}$ is a Hilbert space for each $\alpha \in D$, then $\mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D)$ is a $\mathcal{C}^{\infty}$ -Hilbert manifold modelled on $\prod_{\alpha} W_{\alpha}^{r_{\alpha}} \times \mathbb{R}^{\times r_{\alpha}}$, where $W_{\alpha}^{r_{\alpha}} = (U_{\alpha})^{\perp} \times \cdots$ (each $\mathcal{L}(U_{\alpha}, W_{\alpha}) \cong W_{\alpha}^{r_{\alpha}}$ via the chosen orthonormal basis of U_{α}).*

Example 8.3.5. For $1 < p < \infty$, the L^p tensor space $L^p(\prod_{\alpha} I_{\alpha})$ satisfies the hypotheses of Theorem 8.3.2. Hence $\mathfrak{M}_{\mathbf{t}}((\cdot)_a \otimes_{\alpha} L^p(I_{\alpha}))$ is a $\mathcal{C}^{\infty}$ -Banach manifold. For $p = 2$ it is a Hilbert manifold.

Example 8.3.6. For $V_1 = H^{1,p}(I_1)$, $V_2 = H^{1,p}(I_2)$, and either the full Sobolev norm $\|\cdot\|_{H^{1,p}}$ or the partial norm $\|\cdot\|_{(0,1),p}$ on $V_1 \otimes_a V_2$, both norms make the tensor product map continuous (see [Hac19, Examples 4.41, 4.42]). Hence $\mathfrak{M}_{(r,r)}(H^{1,p}(I_1) \otimes_a H^{1,p}(I_2))$ is a $\mathcal{C}^{\infty}$ -Banach manifold modelled on $\mathcal{L}(U_1, W_1) \times \mathcal{L}(U_2, W_2) \times \mathrm{GL}(\mathbb{R}^r)$.

8.4 The tangent space

Definition 8.4.1 (Tangent subspace $Z^{(D)}(\mathbf{v})$). For $\mathbf{v} \in \mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D)$, define the linear subspace

$$Z^{(D)}(\mathbf{v}) := {}_a \bigotimes_{\alpha \in D} U_{\alpha}^{\min}(\mathbf{v}) \oplus \bigoplus_{\alpha \in D} W_{\alpha}^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) \subset \mathbf{V}_{D, \|\cdot\|_D}. \quad (8.7)$$

$Z^{(D)}(\mathbf{v})$ is a direct sum of $d+1$ subspaces: the core subspace ${}_a \bigotimes_{\alpha} U_{\alpha}^{\min}(\mathbf{v})$ (dimension $\prod r_{\alpha}$) and d “tangent directions” $W_{\alpha}^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$.

Proposition 8.4.2 (Tangent space identification, [FHN19, Proposition 4.13]). *Under the continuity assumption, the tangent map of the inclusion $\mathbf{i} : \mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D, \|\cdot\|_D}$ at $\mathbf{v}$ is injective, and*

$$T_{\mathbf{v}} \mathbf{i}(\mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D))) = Z^{(D)}(\mathbf{v}).$$

More explicitly, for $(\dot{\mathbf{z}}, \dot{C}^{(D)}) \in \mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathbf{t}}(\mathbf{V}_D))$:

$$T_{\mathbf{v}} \mathbf{i}(\dot{\mathbf{z}}, \dot{C}^{(D)}) = \sum_{(i_{\alpha})} \dot{C}_{(i_{\alpha})}^{(D)} \bigotimes_{\alpha} u_{i_{\alpha}}^{(\alpha)} + \sum_{\alpha \in D} \sum_{i_{\alpha}=1}^{r_{\alpha}} \dot{u}_{i_{\alpha}}^{(\alpha)} \otimes \mathbf{U}_{i_{\alpha}}^{(\alpha)}, \quad (8.8)$$

where $\dot{u}_{i_{\alpha}}^{(\alpha)} = \dot{L}_{\alpha}(u_{i_{\alpha}}^{(\alpha)}) \in W_{\alpha}^{\min}(\mathbf{v})$ and $\mathbf{U}_{i_{\alpha}}^{(\alpha)} \in U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ is given by (3.11).

Proof. The formula (8.8) follows by differentiating $(\mathbf{i} \circ \xi_{\mathbf{v}}^{-1})$ at the origin $(\mathbf{o}, C^{(D)})$ using Proposition 3.4.2 and the Leibniz rule for tensor products. Injectivity of $T_{\mathbf{v}}\mathbf{i}$ follows from the linear independence of $\{\otimes_{\alpha} u_{i_{\alpha}}^{(\alpha)}\}$ (a basis of ${}_a \otimes_{\alpha} U_{\alpha}^{\min}(\mathbf{v})$) and of $\{\mathbf{U}_{i_{\alpha}}^{(\alpha)}\}_{i_{\alpha}}$ for each α (a basis of $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$). See [FHN19] for the complete proof. $\square$

8.5 Immersion in the ambient Banach space

To show that $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D, \|\cdot\|_D}$, it remains to show that $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$, i.e., that $Z^{(D)}(\mathbf{v})$ is a split subspace of the ambient Banach space. This requires condition (Inj).

Lemma 8.5.1 ([FHN19, Lemma 4.13]). *Assume (Inj) holds. Let $\beta \in D$ and $W_{\beta} \in \mathbb{G}(V_{\beta, \|\cdot\|_{\beta}})$ with $V_{\beta, \|\cdot\|_{\beta}} = U_{\beta} \oplus W_{\beta}$ for some finite-dimensional U_{β} . Then for every finite-dimensional subspace $U_{[\beta]} \subset \mathbf{V}_{\alpha}^{[\beta]}$:*

$$W_{\beta} \otimes_a U_{[\beta]} \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D}).$$

Proof. The proof uses the continuity of the projection $\text{id}_{\beta} \otimes P_{U_{[\beta]} \oplus W_{[\beta]}}$ on $\mathbf{V}_{D, \|\cdot\|_D}$ (which follows from (Inj) and a norm estimate via [FH12, Lemma 3.18]), and then decomposes $\text{id}_{\beta} = P_{U_{\beta} \oplus W_{\beta}} + P_{W_{\beta} \oplus U_{\beta}}$ to construct an explicit bounded projection onto $W_{\beta} \otimes_a U_{[\beta]}$. See [FHN19] for details. $\square$

Lemma 8.5.2 ([FHN19, Lemma 4.14]). *Let X be a Banach space and $U, V \in \mathbb{G}(X)$ with $U \cap V = \{0\}$. Then $U \oplus V \in \mathbb{G}(X)$ and $U \cap V \in \mathbb{G}(X)$.*

Proof. Since $U, V \in \mathbb{G}(X)$ there exist $U', V' \in \mathbb{G}(X)$ with $X = U \oplus U' = V \oplus V'$. Then $U \oplus V \oplus (U' \cap V') = (U \cap V') \oplus (V \cap U') \oplus (U' \cap V') = X$, proving $U \oplus V \in \mathbb{G}(X)$. The second claim follows from $X = (U \cap V) \oplus (U \cap V') \oplus (V \cap U') \oplus (U' \cap V')$. $\square$

Theorem 8.5.3 (Immersion theorem, [FHN19, Theorem 4.15]). *Assume (Inj) holds. Then for each $\mathbf{v} \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$:*

$$Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D}). \quad (8.9)$$

Consequently, $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D, \|\cdot\|_D}$.

Proof. For each $\alpha \in D$: $W_{\alpha}^{\min}(\mathbf{v}) \in \mathbb{G}(V_{\alpha, \|\cdot\|_{\alpha}})$ and $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ is finite-dimensional. By Lemma 8.5.1, $W_{\alpha}^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$. Also, ${}_a \otimes_{\alpha} U_{\alpha}^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$ (finite-dimensional, hence split by Proposition 2.2.3). The $d+1$ summands in (8.7) have pairwise trivial intersection (by the linear independence properties of minimal subspaces and the direct sum structure), so Lemma 8.5.2 applied iteratively gives $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$. $\square$

Corollary 8.5.4 (Best projection onto tangent space). *If $\mathbf{V}_{D,\|\cdot\|_D}$ is additionally reflexive, then for any $\mathbf{v} \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ and $\dot{\mathbf{u}} \in \mathbf{V}_{D,\|\cdot\|_D}$, there exists $\dot{\mathbf{v}}_{\text{best}} \in Z^{(D)}(\mathbf{v})$ such that*

$$\|\dot{\mathbf{u}} - \dot{\mathbf{v}}_{\text{best}}\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v})} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

Proof. Since $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive, every closed subspace is proximal (proximal = best approximations exist). $Z^{(D)}(\mathbf{v})$ is closed (it is a split subspace by Theorem 8.5.3). $\square$

8.6 The tensor space as a Banach manifold

Corollary 8.6.1 ([FHN19, Corollary 3.19]). *Under the continuity assumption on the tensor product map, the algebraic tensor space $\mathbf{V}_D$ itself is a $\mathcal{C}^\infty$ -Banach manifold (not modelled on a single Banach space), with atlas given by the disjoint union of the Tucker manifold atlases over all $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

$$\mathbf{V}_D = \bigsqcup_{\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D).$$

8.7 Historical notes and comparison

The $\mathcal{C}^\infty$ -Banach manifold structure of $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ in infinite-dimensional spaces was established in [FHN19]. Prior work had treated only finite-dimensional or Hilbert space settings:

Koch and Lubich [KL07] proved that the Tucker manifold is a quotient manifold in finite dimensions and used it for the Dirac–Frenkel variational principle for the MCTDH method. Their approach, however, uses the Hilbert structure (metric projection) and does not extend to Banach spaces.

Uschmajew and Vandereycken [UV13] studied the HT manifold in finite dimensions as a quotient manifold $\text{GL}(r_1) \times \cdots \times \text{GL}(r_d) \backslash \mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) / G$ for a gauge group G . The description in [FHN19] is fundamentally different: it provides an *explicit atlas* with local charts given by formula (8.5), valid in the infinite-dimensional Banach setting.

The key novelties of [FHN19] are:

1. The chart maps $\xi_{\mathbf{v}}$ are given explicitly in terms of the Grassmann chart maps $\Psi_{\mathbf{v}}^{(\alpha)}$, making the geometric structure completely transparent.
2. The immersion theorem (Theorem 8.5.3) holds in general Banach spaces under condition (Inj), without requiring a Hilbert or reflexive structure.
3. The fibre bundle structure identifies the typical fibre as $\mathbb{R}_*^{\times r_\alpha}$, a finite-dimensional smooth manifold (an open subset of a Euclidean space).

Chapter 9

The Tree-Based Manifold

This chapter establishes the geometric counterpart of Chapter 8 for general tree-based tensor formats. The central result is that $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ — when it is a *proper* set of tree-based tensors (Definition 9.1.1) — carries a natural $\mathcal{C}^\infty$ -Banach manifold structure that is compatible with the Tucker manifold structures at each level of the tree, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_D, \|\cdot\|_D$.

The key idea, due to [FHN23], is that $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\text{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ (Theorem 5.4.1) is an intersection of Tucker manifolds, and its chart system is obtained by taking the intersection of the Tucker chart systems at each level. The resulting charts are natural and compatible with those of each Tucker factor.

Throughout, $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces, $\|\cdot\|_D$ satisfies (Inj) at each level of T_D , and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$.

9.1 Proper sets of tree-based tensors

Not every $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ is a genuine intersection of Tucker manifolds — in degenerate cases it may coincide with a single Tucker manifold. The following definition isolates the non-degenerate case.

Definition 9.1.1 (Proper tree-based set). The set $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ is called a *proper set of tree-based tensors* with rank $\mathfrak{r}$ if

$$\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D) \neq \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) \quad \text{for all } 1 \leq k \leq \text{depth}(T_D).$$

Example 9.1.2 (Degenerate case). For $D = \{1, \dots, 6\}$ and the tree T_D with $S(D) = \{\{1\}, \{2, 3, 4, 5, 6\}\}$ (depth 2):

$$\begin{aligned} \mathcal{P}_1(T_D) &= \{\{1\}, \{2, 3, 4, 5, 6\}\}, \\ \mathcal{P}_2(T_D) &= \mathcal{L}(T_D). \end{aligned}$$

Since $\dim U_{\{1\}}^{\min}(\mathbf{v}) = \dim U_{\{2,3,4,5,6\}}^{\min}(\mathbf{v})$ for all $\mathbf{v} \in \mathbf{V}_D$ (Theorem 4.1.3), the constraint at level 1 is automatically satisfied whenever the leaf-level constraints hold. Hence $\mathcal{FT}_\tau(\mathbf{V}_D, T_D) = \mathfrak{M}_{\tau_2}(\mathbf{V}_D, \mathcal{L}(T_D))$ and $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is *not* a proper set. The geometry reduces to the Tucker manifold.

In all examples arising from balanced binary trees (HT format), Tensor Train, and most practical trees with $\text{depth}(T_D) \geq 2$, the set $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is proper.

9.2 The Laplacian-like map and the intersection of model spaces

For each level k of the tree, the Tucker manifold $\mathfrak{M}_{\tau_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ has model space

$$\mathbf{E}_k := \left(\bigoplus_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \otimes \text{span}\{\mathbf{id}_{[\alpha]}\} \right) \times \mathbb{R}^{\times_{\alpha \in \mathcal{P}_k(T_D)} r_\alpha},$$

where $\mathbf{id}_{[\alpha]} = \bigotimes_{\beta \in \mathcal{P}_k(T_D) \setminus \{\alpha\}} \mathbf{id}_{\mathbf{v}_\beta}$ is the identity on all factors except α .

The *Laplacian-like map* Δ_k embeds the product of $\mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$ spaces into $\mathcal{L}(\mathbf{V}_D, \|\cdot\|_D)$:

$$\Delta_k : \prod_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \longrightarrow \mathcal{L}(\mathbf{V}_D, \|\cdot\|_D), \quad (L_\alpha)_\alpha \mapsto \sum_{\alpha \in \mathcal{P}_k(T_D)} L_\alpha \otimes \mathbf{id}_{[\alpha]}. \quad (9.1)$$

Proposition 9.2.1 ([FHN23, Proposition 4.5]). *For $(L_\alpha)_{\alpha \in \mathcal{P}_k(T_D)} \in \prod_{\alpha} \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$:*

$$\exp(\Delta_k((L_\alpha)_\alpha)) = \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \exp(L_\alpha). \quad (9.2)$$

Proof. Put $L = \sum_{\alpha} L_\alpha \otimes \mathbf{id}_{[\alpha]}$. For $\alpha \neq \beta$: $(L_\alpha \otimes \mathbf{id}_{[\alpha]}) \circ (L_\beta \otimes \mathbf{id}_{[\beta]}) = L_\alpha \otimes L_\beta \otimes \mathbf{id}_{[\alpha, \beta]}$, and by the nilpotency of L_α (Proposition 2.2.4), $\exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \exp(L_\alpha) \otimes \mathbf{id}_{[\alpha]}$. Since the summands commute, $\exp(L) = \odot_{\alpha} \exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \bigotimes_{\alpha} \exp(L_\alpha)$. $\square$

The critical consequence is that the local representation $\mathbf{w} = \exp(L)(\mathbf{u})$ for $L \in \mathbf{E}_k$ is *independent of k* : for any k and any $L \in \mathbf{E}_k$ with the same underlying tensor $L = \Delta_k((L_\alpha)_\alpha)$, the formula $\mathbf{w} = (\bigotimes_{\alpha} \exp(L_\alpha))(\mathbf{u})$ gives the same $\mathbf{w}$.

9.3 Charts for the tree-based manifold

Intersection of chart domains

For $\mathbf{v} \in \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$, define the *tree-based chart domain*:

$$\mathcal{W}(\mathbf{v}) := \bigcap_{k=1}^{\text{depth}(T_D)} \mathcal{U}^{(k)}(\mathbf{v}), \quad (9.3)$$

where $\mathcal{U}^{(k)}(\mathbf{v})$ is the chart domain of $\mathfrak{M}_{\tau_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ at $\mathbf{v}$ (Section 8.2).

Model space

Define the *tree-based model space*:

$$\mathbf{E}(\mathbf{v}) := \bigcap_{k=1}^{\text{depth}(T_D)} \mathbf{E}_k \subset \mathcal{L}(\mathbf{V}_D, \|\cdot\|_D), \quad (9.4)$$

and the *tree-based core set*:

$$\mathcal{O}(\mathbf{v}) := \bigcap_{k=1}^{\text{depth}(T_D)} \mathfrak{M}_{\tau_k}(\cdot) \Big|_{\bigotimes_{\alpha \in \mathcal{P}_k(T_D)} U_\alpha^{\min}(\mathbf{v})}.$$

Lemma 9.3.1 ([FHN23, Lemma 4.6]). *Under condition (Inj) at each level, $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ is an open subset of the Banach space $\mathbf{E}(\mathbf{v}) \times \bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$.*

Sketch. Each $\mathcal{O}_k := \mathfrak{M}_{\tau_k}(\cdot) \Big|_{\bigotimes_{\alpha} U_\alpha^{\min}(\mathbf{v})}$ is open in its finite-dimensional ambient space (it is defined by rank conditions, cf. Remark 3.2.3). The intersection $\mathcal{O}(\mathbf{v}) = \bigcap_k \mathcal{O}_k$ is taken in $\bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ (since all factors have the same underlying space by the tree structure), and is an open subset. Each $\mathbf{E}_k$ is a closed subspace of $\mathcal{L}(\mathbf{V}_D, \|\cdot\|_D)$ (Lemma 8.5.1), so $\mathbf{E}(\mathbf{v})$ is a Banach space and $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ is open. $\square$

Tree-based chart map

Since $L \in \mathbf{E}(\mathbf{v})$ satisfies $L \in \mathbf{E}_k$ for all k , the formula $\mathbf{w} = \exp(L)(\mathbf{u})$ is well-defined and independent of the level index k by Proposition 9.2.1. This allows us to define:

Definition 9.3.2 (Tree-based chart). For $\mathbf{v} \in \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$, the *tree-based chart map*

$$\xi_{\mathbf{v}} : \mathcal{W}(\mathbf{v}) \longrightarrow \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v}) \quad (9.5)$$

is defined by $\xi_{\mathbf{v}}(\mathbf{w}) = (L, \mathbf{u})$ where $\mathbf{w} = \exp(L)(\mathbf{u})$, with $L \in \mathbf{E}(\mathbf{v})$ and $\mathbf{u} \in \mathcal{O}(\mathbf{v})$ uniquely determined by any Tucker chart $\xi_{\mathbf{v}}^{(k)}$.

9.4 The $\mathcal{C}^\infty$ -Banach manifold structure

Theorem 9.4.1 ($\mathcal{C}^\infty$ tree-based manifold, [FHN23, Theorem 4.7]). *Let T_D be a dimension partition tree over D with $\text{depth}(T_D) \geq 2$, and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ be such that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a proper set of tree-based tensors. Assume (Inj) holds at each level of T_D . Then:*

- (i) *The collection $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)}$ is a $\mathcal{C}^\infty$ -atlas on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, making it a $\mathcal{C}^\infty$ -Banach manifold modelled on $\mathbf{E}(\mathbf{v}) \times_a \bigotimes_{\alpha \in \mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$.*
- (ii) *The atlas is compatible with the Tucker manifold atlas at each level: the inclusion*

$$\mathbf{i}_k : \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \hookrightarrow \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

is a $\mathcal{C}^\infty$ -immersion for each $1 \leq k \leq \text{depth}(T_D)$.

Sketch. The atlas conditions follow from Lemma 9.3.1 and the fact that the transition maps of the tree-based atlas are restrictions of the transition maps of the Tucker atlases (which are $\mathcal{C}^\infty$ by Lemma 8.3.1). Compatibility with the Tucker atlases is a consequence of the identity $\boldsymbol{\xi}_{\mathbf{v}} = \xi_{\mathbf{v}}^{(k)}|_{\mathcal{W}(\mathbf{v})}$ for each k . See [FHN23] for the complete proof. $\square$

9.5 Tangent space of the tree-based manifold

Proposition 9.5.1 (Tangent space). *Under the assumptions of Theorem 9.4.1, for $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$:*

$$T_{\mathbf{v}}\mathbf{i}\left(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))\right) = \bigcap_{k=1}^{\text{depth}(T_D)} T_{\mathbf{v}}\mathbf{i}_k\left(\mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)))\right) = \bigcap_{k=1}^{\text{depth}(T_D)} Z^{(D)}([]\mathcal{P}_k(T_D)]\mathbf{v}, \quad (9.6)$$

where $Z^{(D)}([]\mathcal{P}]\mathbf{v}$ is the tangent subspace (8.7) for the partition $\mathcal{P}$.

Proof. The model space of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is $\mathbf{E}(\mathbf{v}) \times \mathcal{O}_1$, where $\mathbf{E}(\mathbf{v}) = \bigcap_k \mathbf{E}_k$. Since $T_{\mathbf{v}}\mathbf{i}_k$ is injective (Proposition 8.4.2), the tangent map $T_{\mathbf{v}}\mathbf{i}$ is also injective and its image is $\bigcap_k T_{\mathbf{v}}\mathbf{i}_k(\mathbb{T}_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k()))$. The last equality follows from the explicit formula (8.8) for each level. $\square$

9.6 Immersion in the ambient Banach space

Theorem 9.6.1 (Immersion of the tree-based manifold, [FHN23, Corollary 4.9]). *Under the assumptions of Theorem 9.4.1:*

$$T_{\mathbf{v}}\mathbf{i}\left(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))\right) \in \mathbb{G}(\mathbf{V}_D, \|\cdot\|_D) \quad \text{for all } \mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D). \quad (9.7)$$

Consequently, $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is an immersed submanifold of $\mathbf{V}_{D, \|\cdot\|_D}$.

Proof. From Theorem 8.5.3 applied at each level, $Z^{(D)}(\mathcal{P}_k(\cdot))\mathbf{v} \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$ for $1 \leq k \leq \text{depth}(T_D)$. By Proposition 9.5.1, the tangent space of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is $\bigcap_k Z^{(D)}(\mathcal{P}_k(\cdot))\mathbf{v}$. Applying Lemma 8.5.2 iteratively (intersection of split subspaces is split), we obtain $\bigcap_k Z^{(D)}(\mathcal{P}_k(\cdot))\mathbf{v} \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$. $\square$

Remark 9.6.2 (Embedded vs. immersed). Theorem 9.6.1 shows that $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is an *immersed* submanifold of $\mathbf{V}_{D, \|\cdot\|_D}$. Moreover, Theorem 9.4.1(ii) shows it is an *embedded* submanifold of each Tucker manifold $\mathfrak{M}_{\tau_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$. Whether $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is embedded in $\mathbf{V}_{D, \|\cdot\|_D}$ (i.e., the manifold topology coincides with the subspace topology) is a more delicate question related to the global topology discussed in Section 9.8.

9.7 Comparison with alternative descriptions

The geometric description of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ in [FHN23] differs from earlier approaches in several respects.

Uschmajew–Vandereycken [UV13]. For the HT format in finite-dimensional spaces, they construct $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ as a quotient manifold $P \backslash (\text{GL}(r_\alpha) \times \cdots)$ for a gauge group P . This gives a finite-dimensional manifold and is well-suited for numerical algorithms (Riemannian gradient methods). However, it does not extend to infinite-dimensional spaces and does not make the compatibility with Tucker manifolds at each level explicit.

Holtz–Rohwedder–Schneider [HRS12]. For the TT format in finite dimensions, a similar quotient construction is used. The connection to the Tucker manifold at each level is not made explicit.

Falcó–Hackbusch–Nouy (2015 preprint [FHN15]). An earlier version of the atlas in [FHN23] was proposed using a different chart system. The description in [FHN23] is more natural because the chart maps are directly inherited from the Tucker manifold charts via the Laplacian map (Proposition 9.2.1).

Key novelties of [FHN23].

1. The atlas $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}$ is *natural*: it is the restriction of the Tucker atlases to the intersection $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$, without any additional construction.
2. The *compatibility* with Tucker manifold structures at each level is built into the definition, making the geometric relationships transparent.

3. The description holds in *infinite-dimensional Banach spaces* under the mild condition (Inj).
4. The tangent space formula (9.6) is explicit and computable.

9.8 Topology and connected components

Proposition 9.8.1. *For the Tucker manifold, $\mathfrak{M}_\tau(\mathbf{V}_D)$ is connected for any $\tau \in \mathcal{AD}(\mathbf{V}_D)$ with $\tau \neq \mathbf{0}$. This follows because $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ is connected (each $\mathbb{G}_r(V)$ is connected as a Banach manifold) and the fibre $\mathbb{R}_*^{\times r_\alpha}$ is connected.*

For general tree-based formats, the connectedness of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is an open problem in general. The following is known:

- Proposition 9.8.2.** (i) *For the TT format and binary HT formats in finite dimensions, $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is connected (see [UV13, HRS12]).*
- (ii) *For general T_D and infinite-dimensional spaces, connectedness of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is an open question.*

Remark 9.8.3 (Stratification). The decomposition $\mathbf{V}_D = \sqcup_{\tau \in \mathcal{AD}(\mathbf{V}_D, T_D)} \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ (Theorem 6.6.1) is a stratification of $\mathbf{V}_D$ by tree-based rank. Each stratum $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is an immersed manifold, and $\mathcal{FT}_{\leq \tau}(\mathbf{V}_D, T_D) = \overline{\mathcal{FT}_\tau(\mathbf{V}_D, T_D)}^{\text{weak}}$ is the weak closure. The boundary strata $\partial \mathcal{FT}_{\leq \tau}(\mathbf{V}_D, T_D) = \mathcal{FT}_{\leq \tau}(\mathbf{V}_D, T_D) \setminus \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ consist of tensors with strictly smaller rank. This stratification is the tensor-format analogue of the rank stratification of matrices.

9.9 Historical notes

The geometric structure of $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ as a Banach manifold was established in [FHN23]. The fundamental observation — that the tree-based manifold is an intersection of Tucker manifolds and that its atlas can be obtained by intersecting the Tucker atlases — was made possible by the reformulation of the Tucker manifold charts using the Laplacian-like map Δ_k (Proposition 9.2.1).

The compatibility condition between the tree-based atlas and the Tucker atlases at each level (Theorem 9.4.1(ii)) is the key property that distinguishes the description of [FHN23] from earlier work: it allows one to apply the Tucker manifold geometry (tangent spaces, projections, Dirac–Frenkel principle) at each level of the tree independently, and then combine the results.

The Dirac–Frenkel variational principle for tree-based formats (Chapter 10) exploits this compatibility in an essential way: the reduced ODE on $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is obtained

by projecting the vector field onto the tangent space $\cap_k Z^{(D)}(\cdot)\mathcal{P}_k(\cdot)\mathbf{v}$, and the existence of this projection (Corollary 8.5.4 applied at each level) follows from the fact that each $Z^{(D)}(\cdot)\mathcal{P}_k(\cdot)\mathbf{v}$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$.

Part IV

**Variational Principles and
Applications**

Chapter 10

The Dirac–Frenkel Variational Principle

The geometric framework developed in Chapters 8 and 9 has a direct dynamical application: it provides the rigorous foundation for *model reduction* of differential equations on tensor Banach spaces. Given an abstract Cauchy problem on $\mathbf{V}_{D, \|\cdot\|_D}$, one looks for an approximate solution that evolves on the manifold $\mathfrak{M}_\tau(\mathbf{V}_D)$ (or $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$). The natural way to constrain the dynamics to the manifold is the *Dirac–Frenkel variational principle*: at each time t , the velocity $\dot{\mathbf{v}}_r(t)$ is chosen as the best approximation of the vector field $\mathbf{F}(t, \mathbf{v}_r(t))$ in the tangent space $Z^{(D)}(\mathbf{v}_r(t))$.

This principle was originally proposed by Dirac and Frenkel in 1930 for the time-dependent Schrödinger equation in Hilbert spaces. The present chapter extends it to general Banach tensor spaces, establishing the existence of the reduced ODE and characterising its solutions. Sections 10.1–10.2 set up the abstract framework. Section 10.4 proves the Tucker version (Theorems 5.2 and 5.4 of [FHN19]). Section 10.5 extends to the tree-based setting ([FHN23]).

10.1 The abstract Cauchy problem on tensor Banach spaces

Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces for $\alpha \in D$, and let $\mathbf{V}_{D, \|\cdot\|_D} = \|\cdot\|_D \otimes_{\alpha \in D} V_\alpha$ be a reflexive Banach tensor space satisfying condition (Inj). Consider the abstract Cauchy problem:

$$\dot{\mathbf{u}}(t) = \mathbf{F}(t, \mathbf{u}(t)), \quad t \geq 0, \tag{10.1}$$

$$\mathbf{u}(0) = \mathbf{u}_0, \tag{10.2}$$

where $\mathbf{u}_0 \neq \mathbf{0}$, and $\mathbf{F} : [0, \infty) \times \mathbf{V}_{D, \|\cdot\|_D} \rightarrow \mathbf{V}_{D, \|\cdot\|_D}$ satisfies the usual conditions for existence and uniqueness of solutions (e.g., Lipschitz continuity in the second argument).

The goal is to approximate $\mathbf{u}(t)$ by a *differentiable curve* $t \mapsto \mathbf{v}_r(t)$ lying in $\mathfrak{M}_r(\mathbf{V}_D)$ for some fixed $r \in \mathcal{AD}(\mathbf{V}_D)$ with $r \neq \mathbf{0}$, with initial condition $\mathbf{v}_r(0) = \mathbf{v}_0 \in \mathfrak{M}_r(\mathbf{V}_D)$ chosen as an approximation of $\mathbf{u}_0$.

Remark 10.1.1. By Theorem 4.4.4, a best approximation $\mathbf{v}_0 \in \mathfrak{M}_{\leq r}(\mathbf{V}_D)$ of $\mathbf{u}_0$ exists when $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive. The initial condition $\mathbf{v}_0 \in \mathfrak{M}_r(\mathbf{V}_D)$ (with exact rank r) can always be arranged by taking $\mathbf{v}_0 \in \mathfrak{M}_{r'}(\mathbf{V}_D)$ for some $r' \leq r$ and adjusting the rank.

10.2 The principle in Hilbert spaces

When $\mathbf{V}_{D, \|\cdot\|_D}$ is a Hilbert space with inner product $\langle \cdot, \cdot \rangle_D$, the constraint to $\mathfrak{M}_r(\mathbf{V}_D)$ takes the classical form. Since $Z^{(D)}(\mathbf{v}_r(t)) \in \mathbb{G}(\mathbf{V}_{D, \|\cdot\|_D})$ (Theorem 8.5.3), the orthogonal projection $\mathcal{P}_{\mathbf{v}_r(t)}$ onto $Z^{(D)}(\mathbf{v}_r(t))$ is well-defined and the *Dirac–Frenkel equation* reads:

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t))), \quad (10.3)$$

which is equivalent to the variational condition:

$$\langle \dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t)), \dot{\mathbf{v}} \rangle_D = 0 \quad \text{for all } \dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t)). \quad (10.4)$$

The MCTDH (Multi-Configuration Time-Dependent Hartree) method of quantum dynamics [Lub08] is precisely (10.3) with $\mathbf{V}_{D, \|\cdot\|_D} = L^2(\mathbb{R}^{3d})$ and $\mathfrak{M}_r(\mathbf{V}_D)$ the Hartree manifold ($r = (1, \dots, 1)$).

10.3 Projections in Banach spaces

In a general Banach space, there is no canonical inner product and hence no canonical projection. Two notions replace the orthogonal projection.

Metric projection

Definition 10.3.1. For $\mathbf{v}_r(t) \in \mathfrak{M}_r(\mathbf{V}_D)$ and $t \in I$, the *metric projection* $\mathcal{P}_{\mathbf{v}_r(t)}$ maps $\dot{\mathbf{u}} \in \mathbf{V}_{D, \|\cdot\|_D}$ to the nearest point in $Z^{(D)}(\mathbf{v}_r(t))$:

$$\mathcal{P}_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) := \arg \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

By Corollary 8.5.4, when $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive the metric projection exists. When $\mathbf{V}_{D, \|\cdot\|_D}$ is strictly convex it is unique (since $Z^{(D)}(\mathbf{v}_r(t))$ is convex). The metric projection is, in general, a *nonlinear* operator on Banach spaces.

Normalised duality mapping

The duality mapping $J : \mathbf{V}_{D, \|\cdot\|_D} \rightarrow 2^{\mathbf{V}_{D, \|\cdot\|_D}^*}$ is defined by

$$J(\mathbf{u}) := \left\{ f \in \mathbf{V}_{D, \|\cdot\|_D}^* : \langle \mathbf{u}, f \rangle = \|\mathbf{u}\|_D^2 = \|f\|_*^2 \right\}.$$

It is single-valued when $\mathbf{V}_{D, \|\cdot\|_D}$ is smooth (Definition 2.1.5), and uniformly continuous on bounded sets when $\mathbf{V}_{D, \|\cdot\|_D}$ is uniformly smooth. In a Hilbert space, J coincides with the Riesz isomorphism.

Theorem 10.3.2 (Metric projection characterisation, [FHN19, Theorem 5.2]). *Let $\mathbf{V}_{D, \|\cdot\|_D}$ be reflexive and strictly convex, and assume (Inj). For $\mathbf{v}_r(t) \in \mathfrak{M}_\tau(\mathbf{V}_D)$ and fixed $t \in I$, the following are equivalent:*

- (a) $\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t)))$.
- (b) $\langle \dot{\mathbf{v}}_r(t) - \dot{\mathbf{v}}, J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle \geq 0$ for all $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.
- (c) $\langle \dot{\mathbf{v}}, J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle = 0$ for all $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.

Proof. This is a direct consequence of the classical characterisation of the metric projection onto a closed convex set in a reflexive, strictly convex Banach space, combined with [Alb96, Proposition 2.10]. Condition (c) recovers the Hilbert-space variational form: when J is the Riesz map, (c) becomes the orthogonality condition (10.4). $\square$

Generalised projection

When $\mathbf{V}_{D, \|\cdot\|_D}$ is additionally smooth, an alternative projection is available. Define $\phi : \mathbf{V}_{D, \|\cdot\|_D} \times \mathbf{V}_{D, \|\cdot\|_D} \rightarrow \mathbb{R}$ by

$$\phi(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}\|_D^2 - 2\langle \mathbf{u}, J(\mathbf{v}) \rangle + \|\mathbf{v}\|_D^2.$$

The *generalised projection* $\Pi_{\mathbf{v}_r(t)}$ minimises ϕ over $Z^{(D)}(\mathbf{v}_r(t))$:

$$\Pi_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) := \arg \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \phi(\dot{\mathbf{v}}, \dot{\mathbf{u}}).$$

It coincides with the metric projection when $\mathbf{V}_{D, \|\cdot\|_D}$ is a Hilbert space.

10.4 The Dirac–Frenkel principle for the Tucker manifold

Theorem 10.4.1 (Dirac–Frenkel for Tucker, [FHN19, Theorems 5.2, 5.4]). *Let $\mathbf{V}_{D, \|\cdot\|_D}$ be reflexive and strictly convex, and assume (Inj). Let $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathbf{r} \neq \mathbf{0}$, and let*

$$\mathbf{v}_0 \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D).$$

(Existence) *There exists a differentiable curve $t \mapsto \mathbf{v}_r(t)$ in $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$, defined for t in a neighbourhood of 0, satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t))), \quad \mathbf{v}_r(0) = \mathbf{v}_0. \quad (10.5)$$

(Characterisation) *The curve $t \mapsto \mathbf{v}_r(t)$ satisfies (10.5) if and only if $\dot{\mathbf{v}}_r(t) \in Z^{(D)}(\mathbf{v}_r(t))$ and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \quad (10.6)$$

Sketch. Existence. The reduced ODE (10.5) is an ODE on the Banach manifold $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$. By Theorem 8.5.3, $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_D, \|\cdot\|_D$ for each $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$. The right-hand side of (10.5) depends smoothly on $\mathbf{v}_r(t)$ (the metric projection onto a smoothly-varying split subspace is smooth), so local existence and uniqueness follow from the ODE existence theorem on Banach manifolds [Lan99, Theorem 4.1.5].

Characterisation. Since $Z^{(D)}(\mathbf{v}_r(t))$ is closed and convex, Corollary 8.5.4 gives existence of the minimum in (10.6). The equivalence with the metric projection condition (10.5) follows from the uniqueness (strict convexity) and Theorem 10.3.2. $\square$

Remark 10.4.2 (Kinematic equations). In local chart coordinates $(L_\alpha, C^{(D)})$ at $\mathbf{v}_r(t)$, the reduced ODE (10.5) takes the form of a system of coupled equations for the subspace parameters $L_\alpha(t)$ and the core tensor $C^{(D)}(t)$. The tangent formula (8.8) makes this explicit:

$$\dot{\mathbf{v}}_r(t) = \sum_{(i_\alpha)} \dot{C}_{(i_\alpha)}^{(D)} \bigotimes_{\alpha} u_{i_\alpha}^{(\alpha)} + \sum_{\alpha} \sum_{i_\alpha} \dot{u}_{i_\alpha}^{(\alpha)} \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}.$$

The Dirac–Frenkel condition (10.6) then yields coupled evolution equations for the basis vectors $u_{i_\alpha}^{(\alpha)}(t) \in V_\alpha$ and the core tensor $C^{(D)}(t)$, which generalise the MCTDH equations to arbitrary Tucker ranks in Banach spaces.

The Hartree manifold

Example 10.4.3 (Time-dependent Hartree method). For $\mathbf{r} = \mathbf{1} = (1, \dots, 1)$, the Tucker manifold $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$ consists of elementary tensors $\mathbf{v} = \lambda \bigotimes_{\alpha} v_{\alpha}$ ($v_{\alpha} \in V_{\alpha}$, $\lambda \in \mathbb{R}$), called the *Hartree manifold* [Lub08]. The Dirac–Frenkel ODE (10.5) on the Hartree manifold gives the *time-dependent Hartree equations*. In a Hilbert space with $\|v_{\alpha}\|_{\alpha} = 1$ (unit sphere constraint):

$$\begin{aligned} \dot{\lambda}(t) &= \langle \bigotimes_{\alpha} v_{\alpha}(t), \mathbf{F}(t, \mathbf{v}_r(t)) \rangle_D, \\ \lambda(t) \dot{v}_{\alpha}(t) &= P_{W_{\alpha}(t)} \left[(\text{id}_{\alpha} \otimes \langle \cdot, \bigotimes_{\beta \neq \alpha} v_{\beta}(t) \rangle) \mathbf{F}(t, \mathbf{v}_r(t)) \right], \end{aligned}$$

where $P_{W_\alpha(t)}$ is the orthogonal projection onto the complement of $\text{span}\{v_\alpha(t)\}$ in V_α . This is the system solved numerically by MCTDH.

10.5 Extension to tree-based formats

The extension of Theorem 10.4.1 to $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is immediate given the immersion result of Chapter 9.

Theorem 10.5.1 (Dirac–Frenkel for tree-based formats, [FHN23, Section 5]). *Let T_D be a dimension partition tree, $\tau \in \mathcal{AD}(\mathbf{V}_D, T_D)$ such that $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is a proper set of tree-based tensors. Assume $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive and strictly convex, and that (Inj) holds at each level of T_D . Let $\mathbf{v}_0 \in \mathcal{FT}_\tau(\mathbf{V}_D, T_D)$.*

(Existence) *There exists a differentiable curve $t \mapsto \mathbf{v}_r(t)$ in $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t))), \quad \mathbf{v}_r(0) = \mathbf{v}_0, \quad (10.7)$$

where $\mathcal{P}_{\mathbf{v}_r(t)}$ is the metric projection onto $T_{\mathbf{v}_r(t)}\mathbf{i}(\mathbb{T}_{\mathbf{v}_r(t)}(\mathcal{FT}_\tau(\mathbf{V}_D, T_D))) = \bigcap_k Z^{(D)}([\mathcal{P}_k(T_D)]\mathbf{v}_r(t))$.

(Characterisation) *The curve satisfies (10.7) if and only if $\dot{\mathbf{v}}_r(t) \in \bigcap_k Z^{(D)}([\mathcal{P}_k(T_D)]\mathbf{v}_r(t))$ and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in \bigcap_k Z^{(D)}([\mathcal{P}_k(T_D)]\mathbf{v}_r(t))} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \quad (10.8)$$

Proof. The proof is identical to that of Theorem 10.4.1, with $\mathfrak{M}_\tau(\mathbf{V}_D)$ replaced by $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ and $Z^{(D)}(\mathbf{v}_r(t))$ replaced by $\bigcap_k Z^{(D)}([\mathcal{P}_k(T_D)]\mathbf{v}_r(t))$. The key inputs are: (i) $\mathcal{FT}_\tau(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$ -Banach manifold (Theorem 9.4.1); (ii) its tangent space is a split subspace of $\mathbf{V}_{D, \|\cdot\|_D}$ (Theorem 9.6.1); and (iii) the metric projection onto a split subspace in a reflexive strictly convex space is well-defined and unique. $\square$

Remark 10.5.2 (Algorithmic interpretation). Theorem 10.5.1 provides the mathematical foundation for a class of algorithms that integrate tensor-format ODEs by evolving the transfer tensors $\{C^{(\alpha)}\}_{\alpha \in T_D}$ and the leaf basis vectors $\{u_{i_j}^{(j)}\}_{j \in \mathcal{L}(T_D)}$ along the Dirac–Frenkel trajectories. For the TT format, this is the starting point of the *projector-splitting integrator* of Lubich, Oseledets, and Vandereycken [LOV15]. For general tree-based formats, the corresponding algorithm was introduced by Ceruti, Lubich, and Walach [CLW21].

10.6 Historical notes

The Dirac–Frenkel variational principle was proposed independently by Dirac [Dir30] and Frenkel [Fre34] in 1930 and 1934 for the time-dependent Schrödinger equation in

Hilbert space. The formulation used here — projection of the vector field onto the tangent space of a manifold — is that of McLachlan et al. and Lubich.

Koch and Lubich [KL07] proved existence and uniqueness of solutions to the Dirac–Frenkel ODE on Tucker manifolds in finite-dimensional Hilbert spaces, and used it to analyse the MCTDH method. The extension to infinite-dimensional Banach spaces (Theorem 10.4.1) was established in [FHN19], where the key difficulty — showing that the tangent space $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$, without the Hilbert space structure — was resolved by the immersion theorem (Theorem 8.5.3).

The extension to tree-based formats (Theorem 10.5.1) is the main application result of [FHN23]: having established the geometry of $\mathcal{FT}_r(\mathbf{V}_D, T_D)$ in Chapter 9, the Dirac–Frenkel principle follows with no additional effort.

Chapter 11

Open Problems and Perspectives

The results established in this monograph open several directions for further research. This chapter collects the most natural open problems, organised by theme, together with a discussion of connections to other active areas of mathematics and its applications.

11.1 Global topology of tensor manifolds

Connectedness of tree-based manifolds

We showed in Proposition 9.8.1 that each Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is connected. For tree-based manifolds with $\text{depth}(T_D) \geq 2$, this question is open in the infinite-dimensional setting.

Open Problem 11.1.1. Let T_D be a dimension partition tree with $\text{depth}(T_D) \geq 2$ and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ a proper set. Is $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ connected?

In finite dimensions, connectedness has been established for the HT format [UV13] and the TT format [HRS12], but the methods used there (algebraic geometry of determinantal varieties) do not directly extend to the Banach setting.

Higher homotopy groups

Even in finite dimensions, the homotopy type of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is not fully understood. Since $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a fibre bundle over the product of Grassmannians $\prod_{\alpha} \mathbb{G}_{r_{\alpha}}(V_{\alpha})$ (Theorem 8.3.2), the long exact sequence of homotopy groups relates $\pi_n(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$ to $\pi_n(\prod_{\alpha} \mathbb{G}_{r_{\alpha}}(\mathbb{R}^{n_{\alpha}}))$ and $\pi_n(\mathbb{R}_*^{\times r_{\alpha}})$.

Open Problem 11.1.2. Compute the fundamental group $\pi_1(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$ and $\pi_1(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))$ in the finite-dimensional case. Are these manifolds simply connected?

Rank stratification

The decomposition $\mathbf{V}_D = \bigsqcup_{\mathbf{r}} \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is a stratification of the algebraic tensor space. The boundary $\overline{\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)} \setminus \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ consists of tensors with strictly smaller rank, and the closure $\overline{\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)}^{\|\cdot\|}$ in the Banach topology is the set $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$ of tensors of bounded rank (Theorem 4.4.3).

Open Problem 11.1.3. Describe the local geometry of the boundary strata: what is the tangent cone to $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$ at a boundary point $\mathbf{v} \in \mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D) \setminus \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$? Is the stratification Whitney regular?

11.2 Riemannian and Finsler geometry on tensor manifolds

In the Hilbert case, the Tucker manifold inherits a Riemannian metric from the ambient Hilbert tensor space. Riemannian gradient methods on $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ (ALS, Riemannian gradient, retraction-based methods) have been developed by Uschmajew and Vandereycken [UV20]. In the Banach setting, a natural replacement is a Finsler metric (a continuously varying norm on each tangent space).

Open Problem 11.2.1. Equip $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ with the Finsler metric induced by $\|\cdot\|_D$ (via the isomorphism $T_{\mathbf{v}}\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D) \cong Z^{(D)}(\mathbf{v})$). Study the geodesics, curvature, and exponential map of this Finsler manifold. In particular: does the Dirac–Frenkel flow (10.5) preserve the Finsler structure in any sense?

11.3 Algorithms and convergence analysis

Projector-splitting integrators on tree-based manifolds

For the TT format, Lubich, Oseledets, and Vandereycken [LOV15] introduced the *projector-splitting integrator*, which integrates the Dirac–Frenkel ODE by splitting the projection onto $Z^{(D)}(\mathbf{v}_r(t))$ into a sequence of rank-one updates. For general tree-based formats, the corresponding integrator was introduced in [CLW21].

The convergence analysis of these algorithms in infinite-dimensional Banach spaces (as opposed to Hilbert spaces) requires understanding the stability of the Dirac–Frenkel flow on $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$ and the approximation properties of the splitting.

Open Problem 11.3.1. Extend the convergence analysis of the projector-splitting integrator to the setting of Banach tensor spaces. In particular, establish error bounds that depend explicitly on the geometry of the tree-based manifold $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$ (curvature, injectivity radius, condition number of the rank map $\varrho_{\mathbf{r}}$).

Alternating least squares in Banach spaces

The *alternating least squares* (ALS) algorithm and its variants (DMRG, HOSVD) minimise a functional over $\mathfrak{M}_{\leq \tau}(\mathbf{V}_D)$ by alternating optimisation over one factor at a time. In Hilbert spaces, convergence of ALS is well understood [UV20]. In Banach spaces, convergence requires different tools.

Open Problem 11.3.2. Prove convergence of ALS for minimisation over $\mathfrak{M}_{\leq \tau}(\mathbf{V}_D)$ in a reflexive Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D}$. What role does the geometry of the Tucker manifold (specifically, the structure of the tangent spaces $Z^{(D)}(\mathbf{v})$) play in the convergence rate?

11.4 Beyond Banach spaces

Quasi-Banach and Fréchet spaces

The component spaces V_α in this monograph are normed (Banach or at least normed). Many function spaces of interest — L^p for $0 < p < 1$, or spaces of smooth functions — are quasi-Banach or Fréchet but not Banach. The theory of minimal subspaces and Tucker manifolds should extend to these settings, but the proofs require substantial modifications.

Open Problem 11.4.1. Develop the theory of minimal subspaces and tensor manifolds for quasi-Banach spaces V_α with $0 < p < 1$. In particular, does the Tucker manifold $\mathfrak{M}_\tau(\mathbf{V}_D)$ retain its smooth structure when the V_α are L^p spaces for $p < 1$?

Infinite-dimensional index sets

All results in this monograph assume $D = \{1, \dots, d\}$ is finite. For problems in infinite dimensions — e.g., infinite particle quantum systems or PDEs in infinitely many variables — one would want D to be infinite. The algebraic tensor product over an infinite index set requires projective limits or nuclear spaces.

Open Problem 11.4.2. Extend the tree-based tensor framework to infinite index sets $D = \mathbb{N}$ (or more general sets). What is the natural analogue of a dimension partition tree, and what conditions on the norms guarantee the existence of minimal subspaces and best approximations?

11.5 Connections with quantum information

Tensor networks and entanglement

In quantum information, a *tensor network state* on a lattice is a tensor $\mathbf{v} \in \bigotimes_{j \in D} \mathbb{C}^{d_j}$ whose transfer tensors $\{C^{(\alpha)}\}$ are specified by a graph (not necessarily a tree). The tree-based format corresponds to *tree tensor networks* (TTN), which include MPS/TT as the linear-chain case and MERA (multi-scale entanglement renormalisation ansatz) as a more general tree.

The *entanglement entropy* of a bipartition $D = \alpha \cup \alpha^c$ is $S(\alpha) = -\text{tr}(\rho_\alpha \log \rho_\alpha)$, where ρ_α is the reduced density matrix of the state on α . The *area law* for ground states of local Hamiltonians states that $S(\alpha)$ grows at most as $|\partial\alpha|$ (the boundary of α), not as $|\alpha|$ (the volume). This motivates the use of tree-based formats: the rank constraints in $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$ directly control the entanglement structure.

Open Problem 11.5.1. Formulate the Dirac–Frenkel variational principle for tree tensor networks in the language of C^* -algebras and operator systems. What is the correct generalisation of the Tucker manifold in the non-commutative setting?

11.6 Learning on tensor manifolds

Statistical learning theory

In machine learning, tensor network states arise naturally as *function approximation schemes* for high-dimensional data. The generalisation capacity of these models depends on the geometry of the hypothesis class $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$.

The *Rademacher complexity* of $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$ controls the generalisation error via the Dudley entropy integral. For finite D and fixed ranks, this can be bounded using the Weyl tube formula once the curvature of the manifold $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D)$ is understood.

Open Problem 11.6.1. Compute the Rademacher complexity and metric entropy of $\mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D)$ in terms of the tree-based rank $\mathbf{r}$, the depth $\text{depth}(T_D)$, and the dimensions $\dim V_j$. How does the choice of tree T_D affect the approximation-complexity tradeoff?

Optimal tree selection

Given a function $f : \prod_j I_j \rightarrow \mathbb{R}$ (a target function for approximation), which tree T_D gives the best approximation of f by a tree-based tensor of a given rank? This question connects the geometry of tensor manifolds to the combinatorics of tree structures.

Open Problem 11.6.2. Given $f \in \mathbf{V}_{D, \|\cdot\|_D}$ and a bound r on the tree-based rank, find the dimension partition tree T_D^* that minimises the approximation error $\inf_{\mathbf{v} \in \mathcal{FT}_{\leq r}(\mathbf{V}_D, T_D^*)} \|f - \mathbf{v}\|_D$. Is this optimisation problem tractable?

Appendix A

Banach Manifolds: A Summary

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Appendix B

Functional Analysis: Selected Results

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Appendix C

Index of Notation

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Bibliography

- [Alb96] Ya. I. Alber. *Metric and Generalized Projection Operators in Banach Spaces*. Dekker, New York, 1996.
- [CLW21] Gianluca Ceruti, Christian Lubich, and Hanna Walach. Time integration of tree tensor networks. *SIAM Journal on Numerical Analysis*, 59(1):289–313, 2021.
- [Dir30] Paul A. M. Dirac. *The Principles of Quantum Mechanics*. Oxford University Press, 1930.
- [DLDMV00] Lieven De Lathauwer, Bart De Moor, and Joos Vandewalle. A multilinear singular value decomposition. *SIAM Journal on Matrix Analysis and Applications*, 21(4):1253–1278, 2000.
- [Dou66] Adrien Douady. Le problème des modules pour les sous-espaces analytiques compacts d’un espace analytique donné. *Annales de l’Institut Fourier*, 16(1):1–95, 1966.
- [DR63] Régine Douady and Philippe Robba. Compléments et sous-espaces scindés des espaces de Banach. *Bull. Soc. Math. France*, 91:117–136, 1963.
- [FH12] Antonio Falcó and Wolfgang Hackbusch. On minimal subspaces in tensor representations. *Foundations of Computational Mathematics*, 12(6):765–803, 2012.
- [FHHM15] Antonio Falcó, Wolfgang Hackbusch, Mathieu Herrmann, and Laurent Mouysset. Geometric structures in tensor representations. 2015. arXiv:1505.03027.
- [FHN15] Antonio Falcó, Wolfgang Hackbusch, and Anthony Nouy. Geometric structures in tensor representations. 2015. arXiv:1505.03027.
- [FHN19] Antonio Falcó, Wolfgang Hackbusch, and Anthony Nouy. On the Dirac–Frenkel variational principle on tensor Banach spaces. *Foundations of Computational Mathematics*, 19(1):159–204, 2019.

- [FHN21] Antonio Falcó, Wolfgang Hackbusch, and Anthony Nouy. Tree-based tensor formats. *SeMA Journal*, 78(1):159–173, 2021.
- [FHN23] Antonio Falcó, Wolfgang Hackbusch, and Anthony Nouy. Geometry of tree-based tensor formats in tensor Banach spaces. *Annali di Matematica Pura ed Applicata*, 2023.
- [Fre34] Jakov I. Frenkel. *Wave Mechanics: Advanced General Theory*. Oxford University Press, 1934.
- [Gre81] Werner H. Greub. *Linear Algebra*, volume 23 of *Graduate Texts in Mathematics*. Springer, New York, 4th edition, 1981.
- [Hac19] Wolfgang Hackbusch. *Tensor Spaces and Numerical Tensor Calculus*, volume 56 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2nd edition, 2019.
- [Hit27] Frank L. Hitchcock. The expression of a tensor or a polyadic as a sum of products. *Journal of Mathematics and Physics*, 6:164–189, 1927.
- [HK09] Wolfgang Hackbusch and Stefan Kühn. A new scheme for the tensor representation. *Journal of Fourier Analysis and Applications*, 15(5):706–722, 2009.
- [HN12] Joachim Hilgert and Karl-Hermann Neeb. *Structure and Geometry of Lie Groups*. Springer, New York, 2012.
- [HRS12] Sebastian Holtz, Thorsten Rohwedder, and Reinhold Schneider. The alternating linear scheme for tensor optimization in the tensor train format. *SIAM Journal on Scientific Computing*, 34(2):A683–A713, 2012.
- [KB09] Tamara G. Kolda and Brett W. Bader. Tensor decompositions and applications. *SIAM Review*, 51(3):455–500, 2009.
- [KL07] Othmar Koch and Christian Lubich. Dynamical low-rank approximation. *SIAM Journal on Matrix Analysis and Applications*, 29(2):434–454, 2007.
- [Lan99] Serge Lang. *Fundamentals of Differential Geometry*, volume 191 of *Graduate Texts in Mathematics*. Springer, New York, 1999.
- [LOV15] Christian Lubich, Ivan V. Oseledets, and Bart Vandereycken. Time integration of tensor trains. *SIAM Journal on Numerical Analysis*, 53(2):917–941, 2015.

- [Lub08] Christian Lubich. *From Quantum to Classical Molecular Dynamics: Reduced Models and Numerical Analysis*. Zurich Lectures in Advanced Mathematics. European Mathematical Society, Zürich, 2008.
- [Nou17] Anthony Nouy. Low-rank methods for high-dimensional approximation and model order reduction. In Peter Benner, Albert Cohen, Mario Ohlberger, and Karen Willcox, editors, *Model Reduction and Approximation: Theory and Algorithms*, pages 171–226. SIAM, Philadelphia, 2017.
- [Ose11] Ivan V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5):2295–2317, 2011.
- [Tuc66] Ledyard R. Tucker. Some mathematical notes on three-mode factor analysis. *Psychometrika*, 31(3):279–311, 1966.
- [Upm85] Harald Upmeyer. *Symmetric Banach Manifolds and Jordan C^* -Algebras*. North-Holland, Amsterdam, 1985.
- [UV13] André Uschmajew and Bart Vandereycken. The geometry of algorithms using hierarchical tensors. *Linear Algebra and its Applications*, 439(1):133–166, 2013.
- [UV20] André Uschmajew and Bart Vandereycken. Geometric methods on low-rank matrix and tensor manifolds. pages 261–313, 2020.
- [Vid03] Guifré Vidal. Efficient classical simulation of slightly entangled quantum computations. *Physical Review Letters*, 91(14):147902, 2003.